



US 20250281285A1

(19) **United States**

(12) **Patent Application Publication**
Braido et al.

(10) **Pub. No.: US 2025/0281285 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **COMMISSURE ATTACHMENT FEATURE
HAVING MULTIPLE BODIES**

is a continuation of application No. 13/781,201, filed on Feb. 28, 2013, now abandoned.

(71) Applicant: **St. Jude Medical, Cardiology
Division, Inc.**, St. Paul, MN (US)

(60) Provisional application No. 61/666,224, filed on Jun. 29, 2012.

(72) Inventors: **Peter Nicholas Braido**, Wyoming, MN (US); **Jacob John Daly**, Blaine, MN (US); **Julia Ann Schraut**, Shoreview, MN (US)

Publication Classification

(51) **Int. Cl.**
A61F 2/24 (2006.01)

(52) **U.S. Cl.**
CPC **A61F 2/2418** (2013.01); **A61F 2/2409** (2013.01); **A61F 2/2412** (2013.01); **A61F 2/2415** (2013.01); **A61F 2220/005** (2013.01); **A61F 2220/0075** (2013.01); **A61F 2230/0054** (2013.01); **A61F 2230/0067** (2013.01); **A61F 2230/0069** (2013.01)

(73) Assignee: **St. Jude Medical, Cardiology
Division, Inc.**, St. Paul, MN (US)

(21) Appl. No.: **19/214,814**

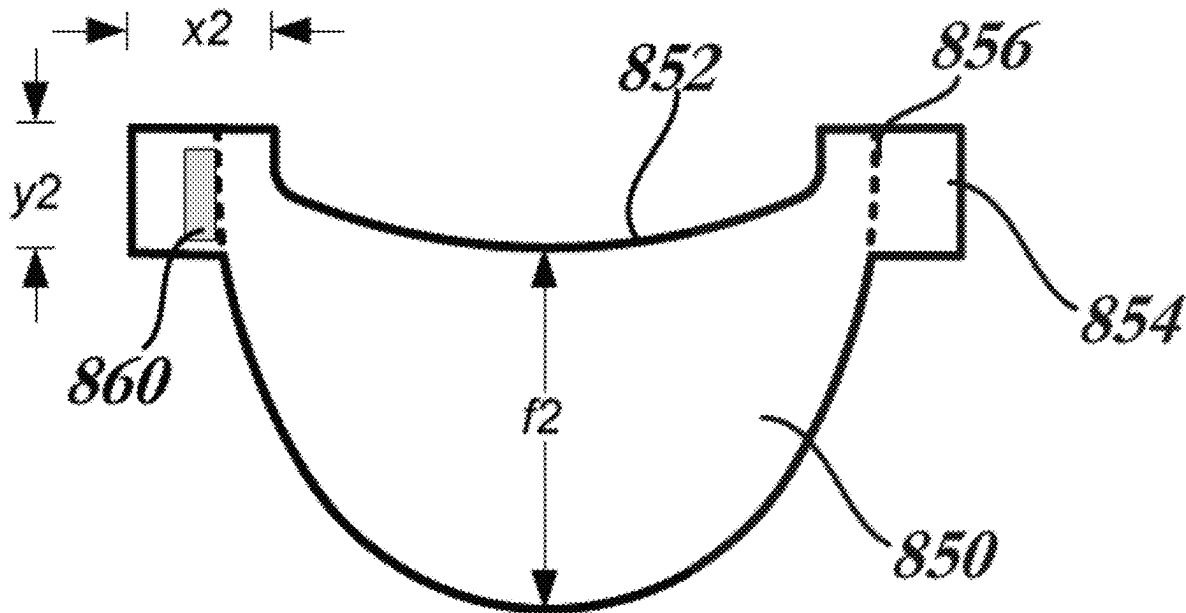
(22) Filed: **May 21, 2025**

Related U.S. Application Data

(60) Division of application No. 17/871,432, filed on Jul. 22, 2022, which is a division of application No. 17/144,248, filed on Jan. 8, 2021, now Pat. No. 11,426,275, which is a division of application No. 15/613,688, filed on Jun. 5, 2017, now abandoned, which is a continuation of application No. 15/202,939, filed on Jul. 6, 2016, now Pat. No. 9,693,861, which is a continuation of application No. 14/824,551, filed on Aug. 12, 2015, now abandoned, which

(57) **ABSTRACT**

A prosthetic heart valve includes a collapsible and expandable stent extending in a longitudinal direction between an inflow end and an outflow end, the stent including a plurality of struts defining a plurality of cells, and a plurality of commissure attachment features disposed on the stent, each commissure attachment feature having a first end, a second end opposite the first end, and at least two bodies connected via one or more suspension struts, each of the at least two bodies having at least one eyelet.



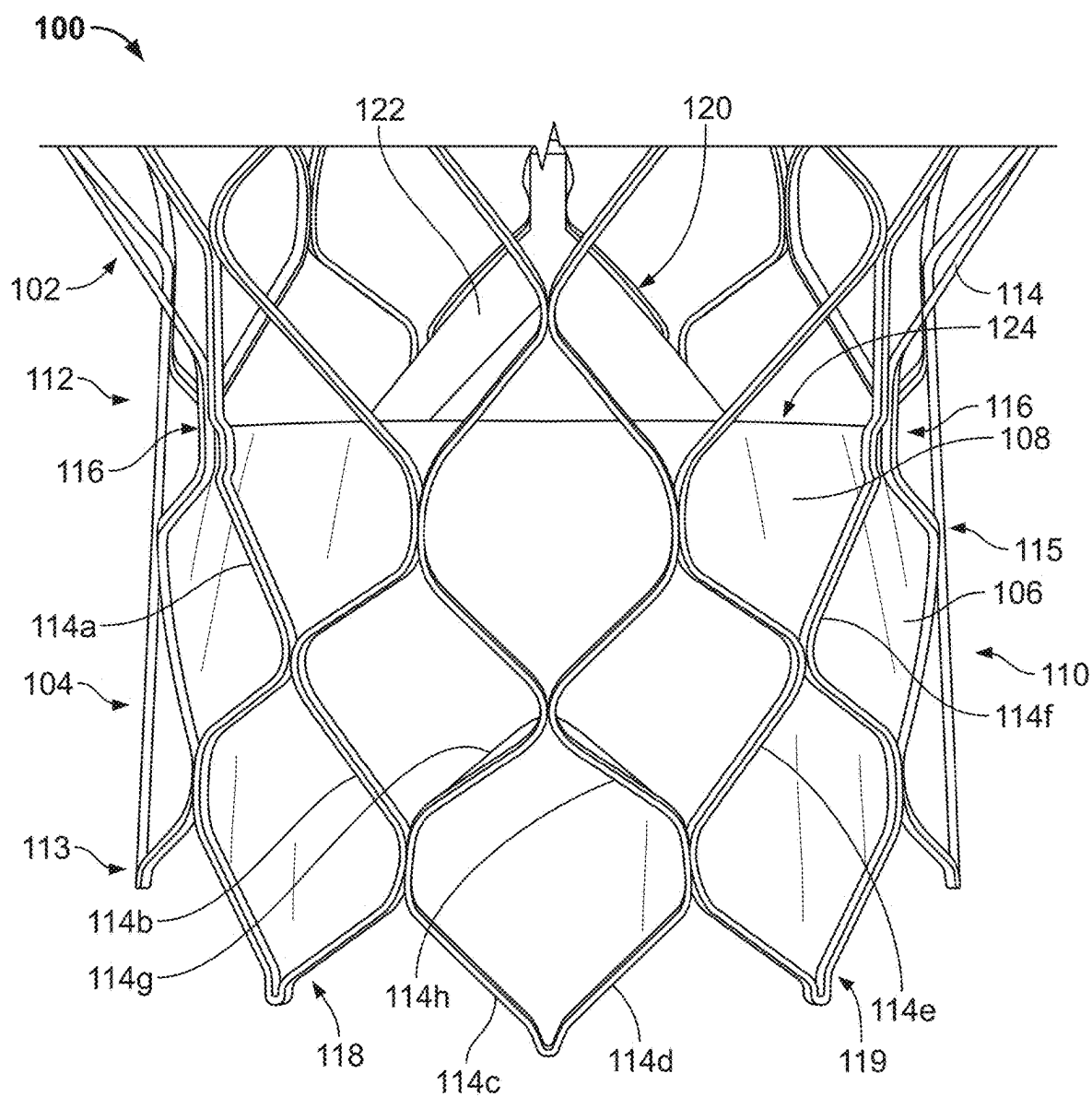


FIG. 1

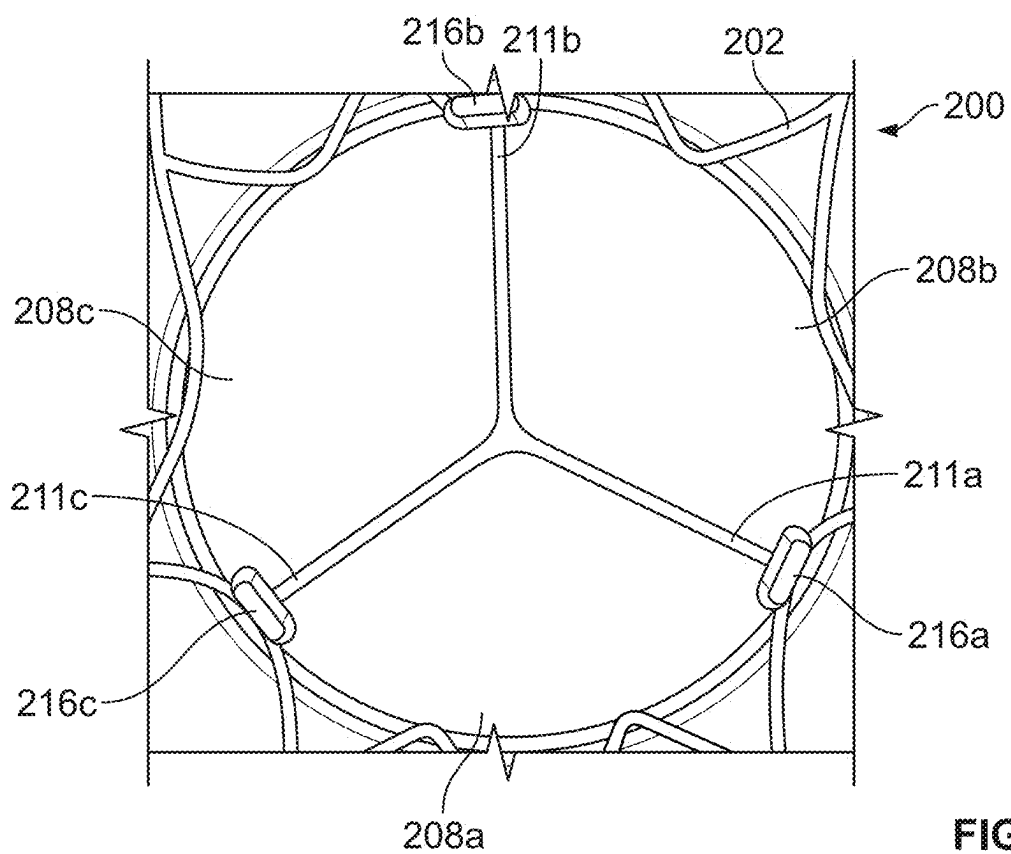


FIG. 2A

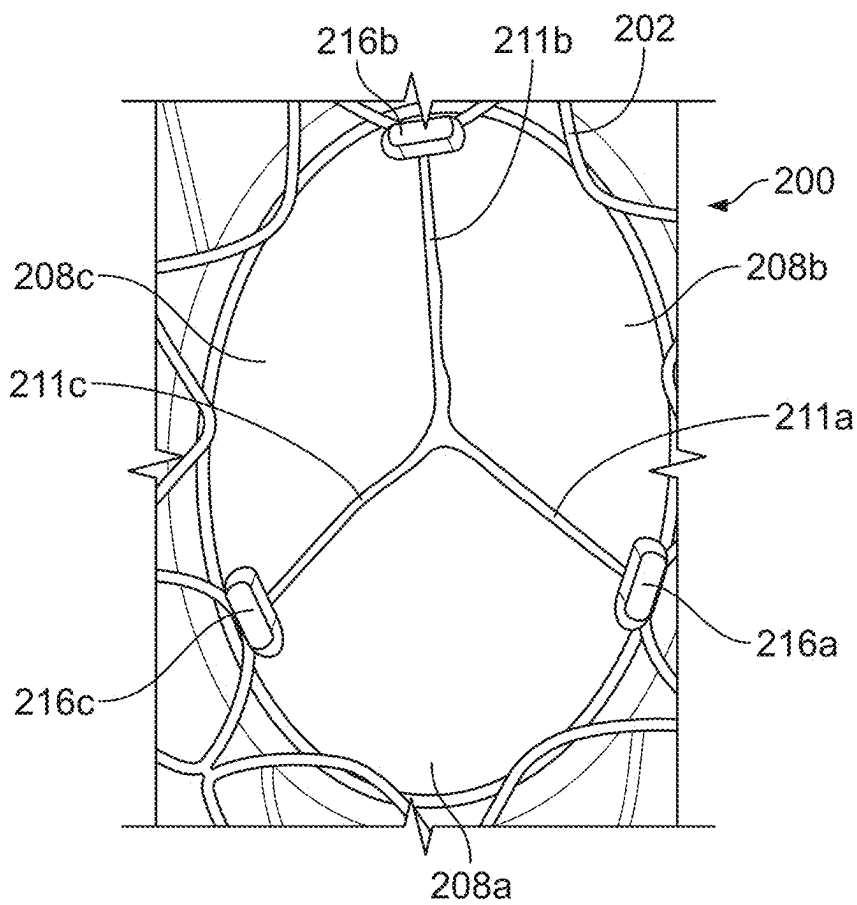


FIG. 2B

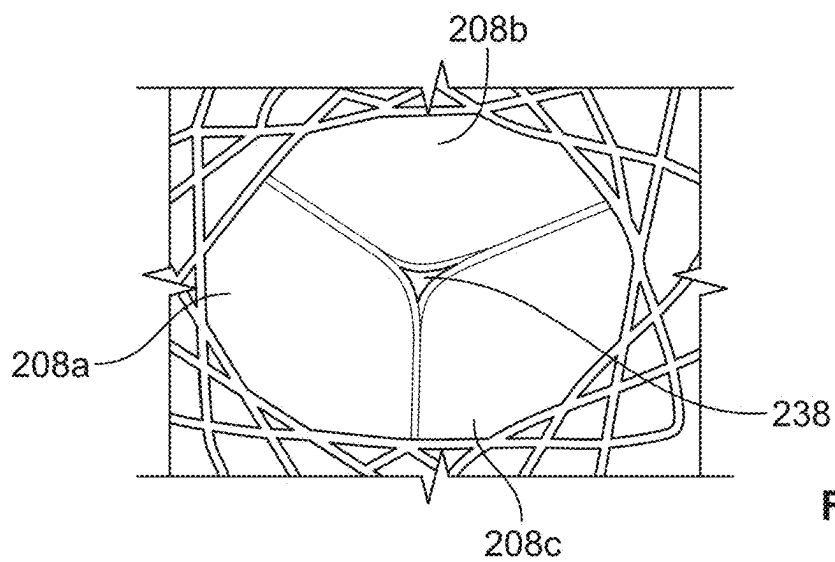


FIG. 3A

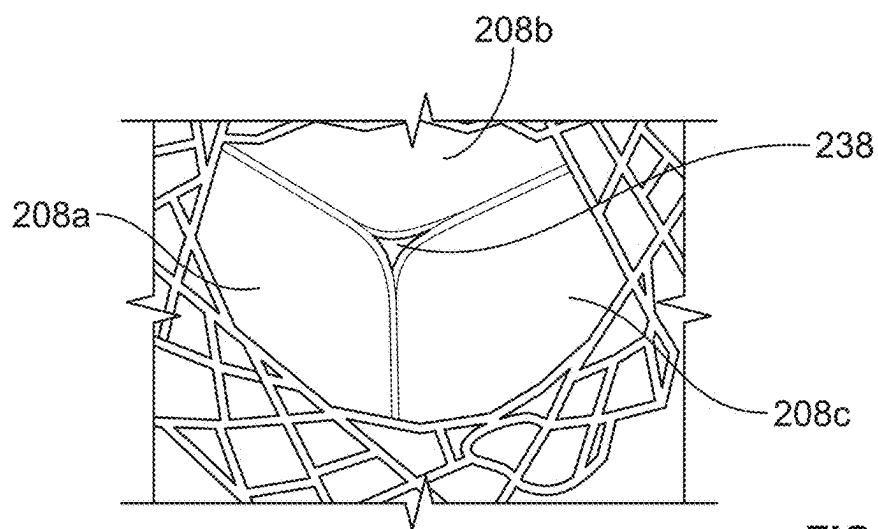


FIG. 3B

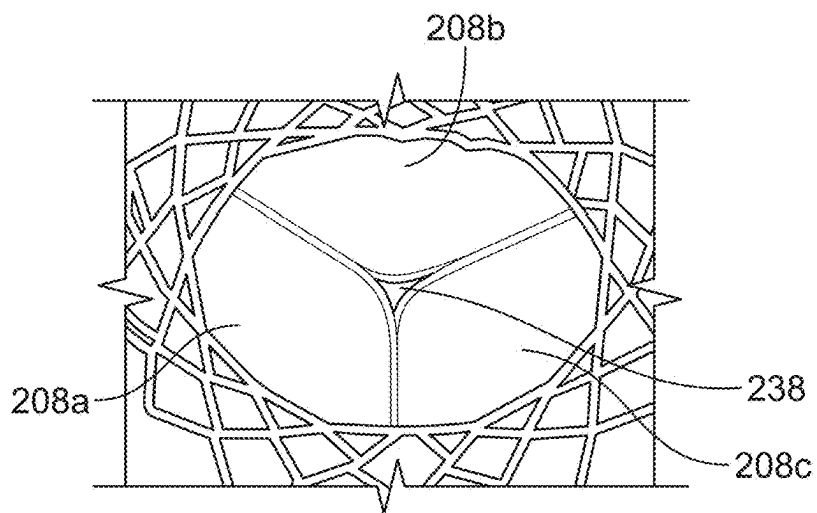


FIG. 3C

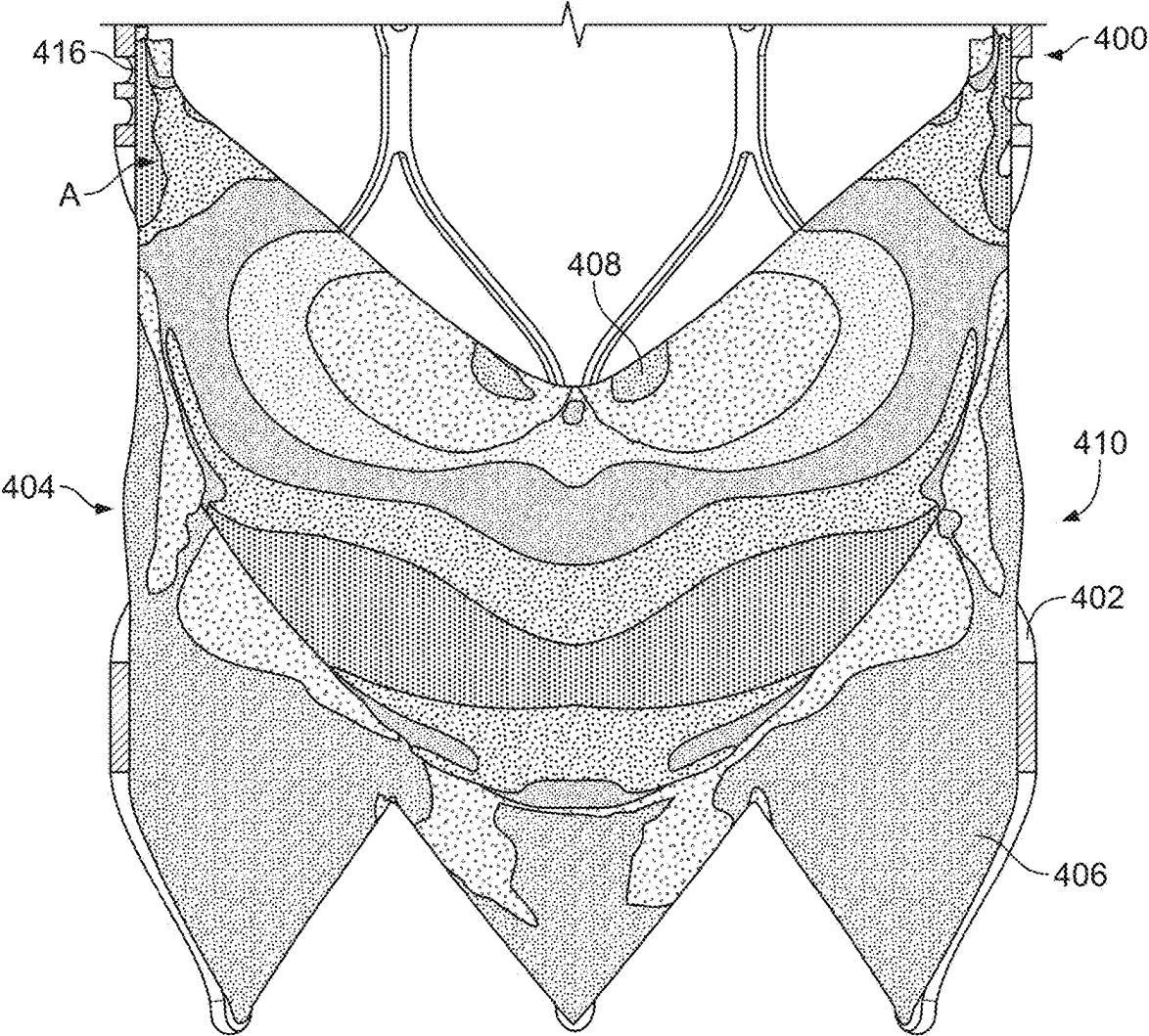


FIG. 4

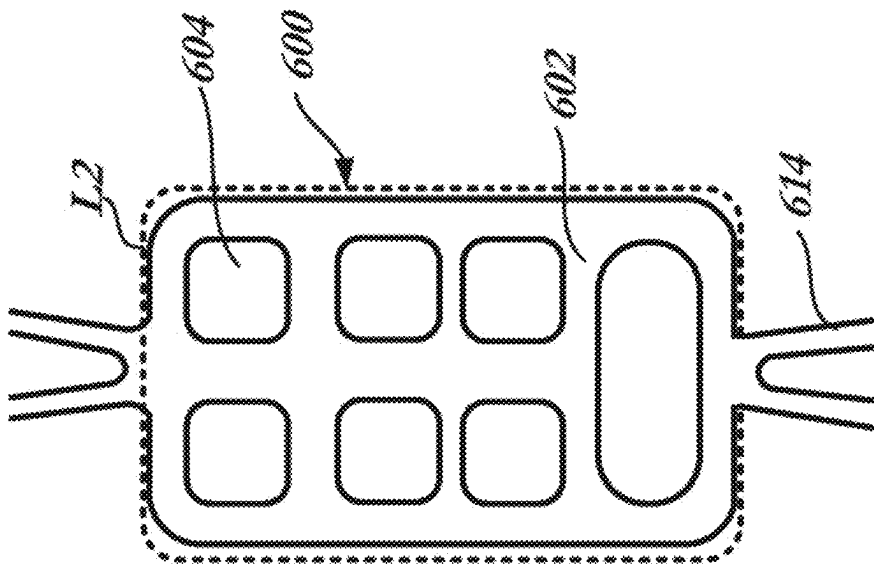


FIG. 5B

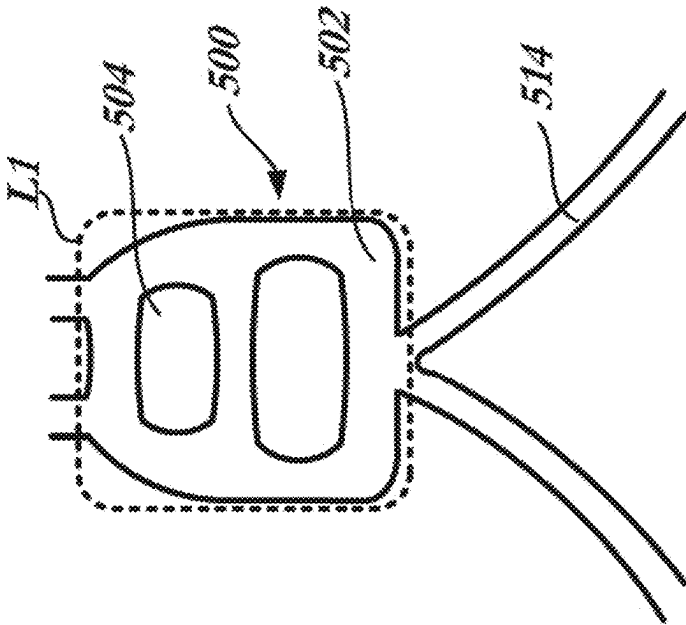
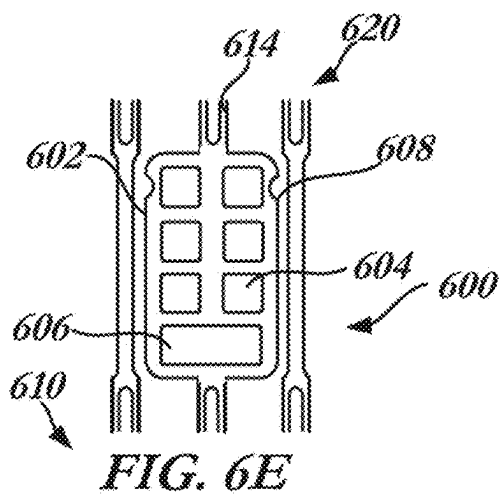
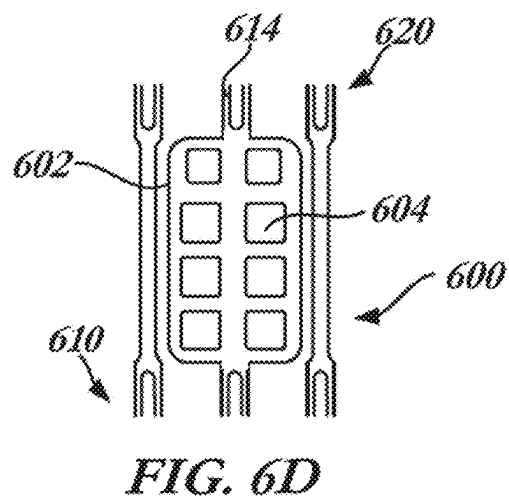
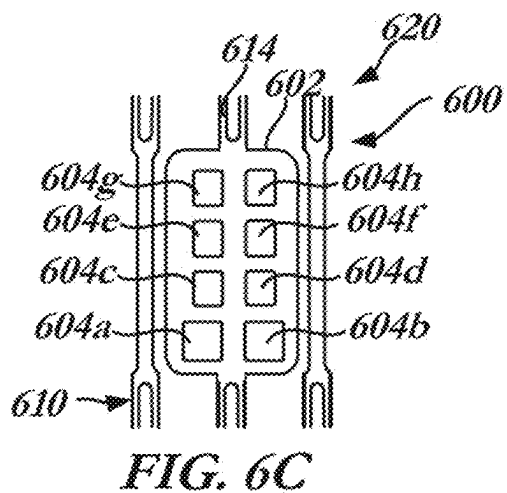
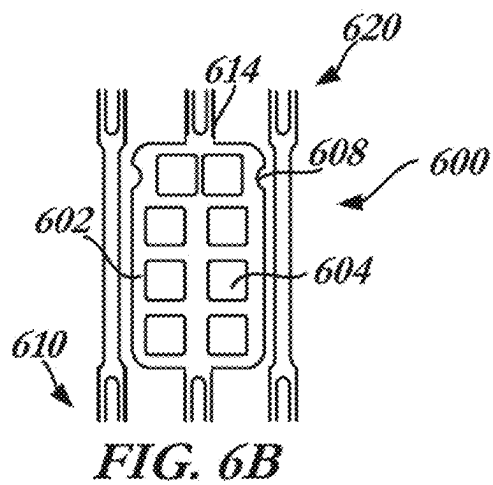
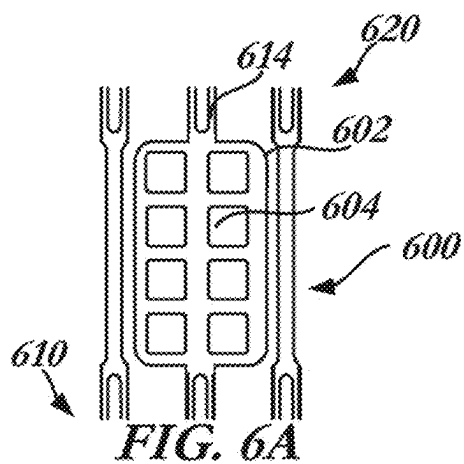
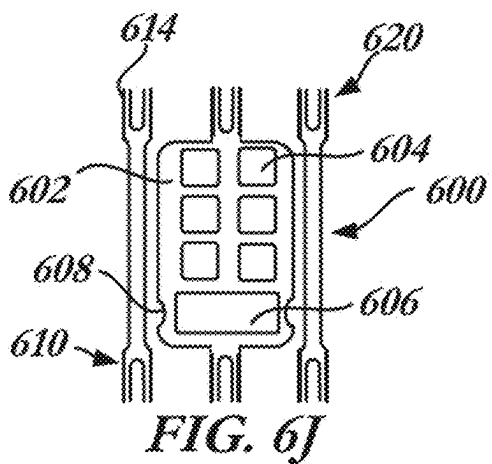
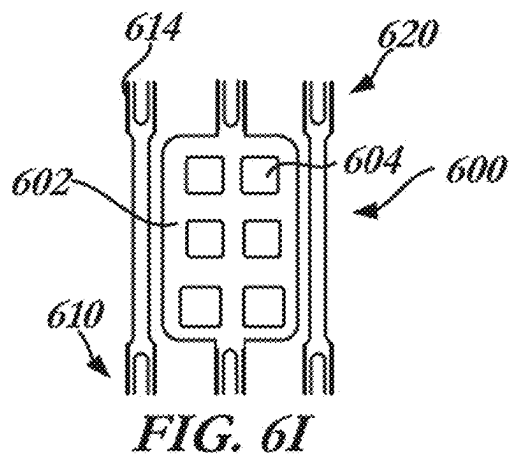
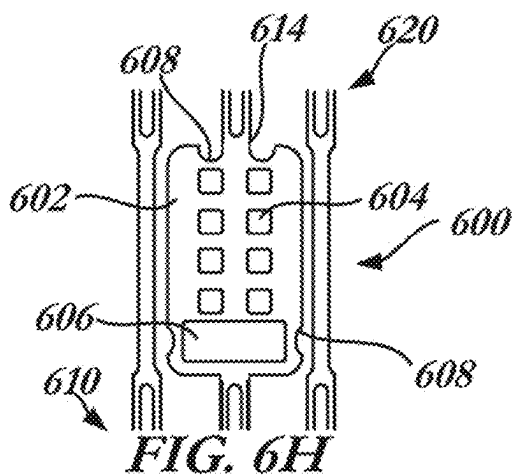
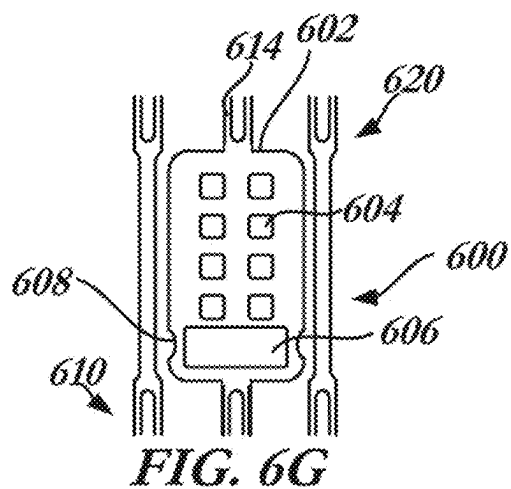
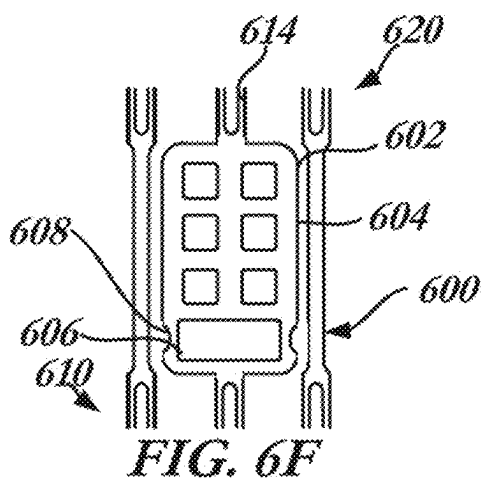


FIG. 5A





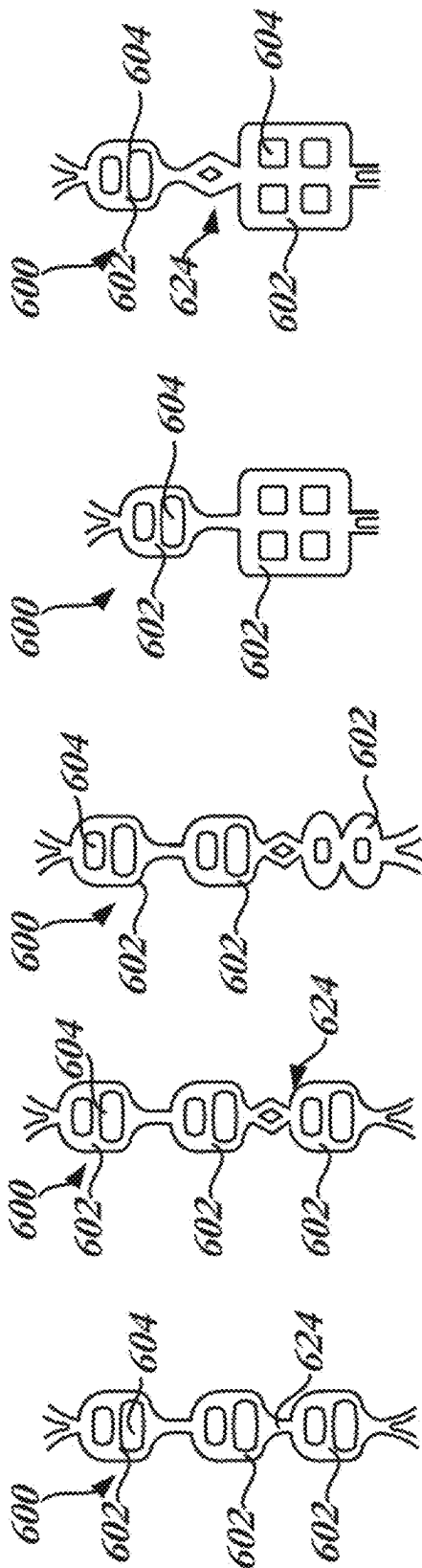


FIG. 6K FIG. 6L FIG. 6M FIG. 6N

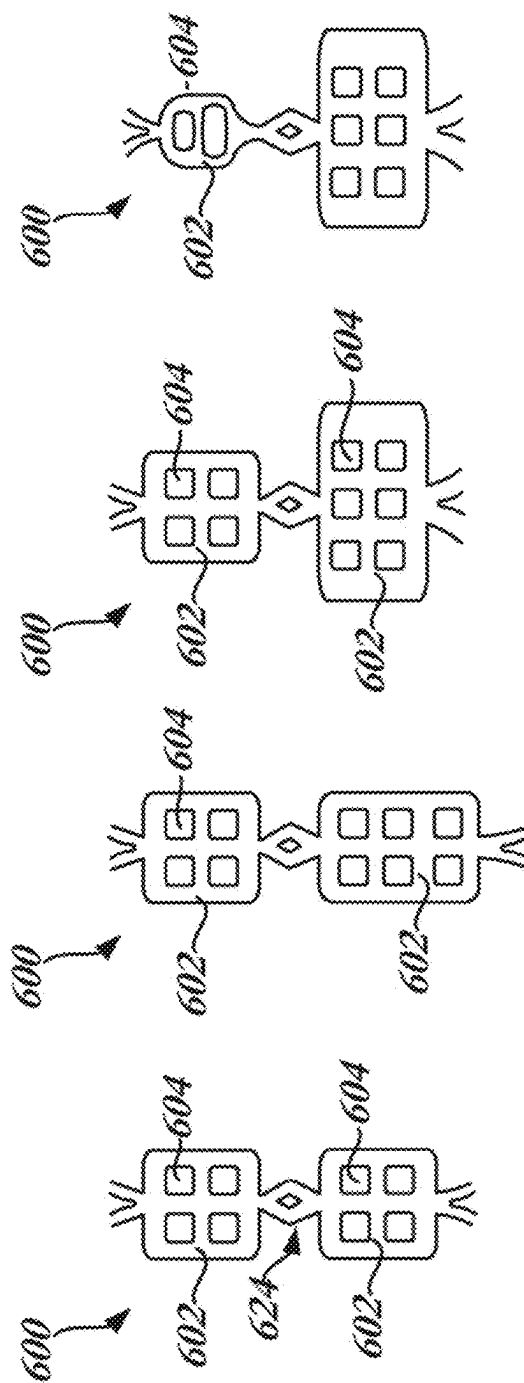


FIG. 6P FIG. 6Q FIG. 6R FIG. 6S

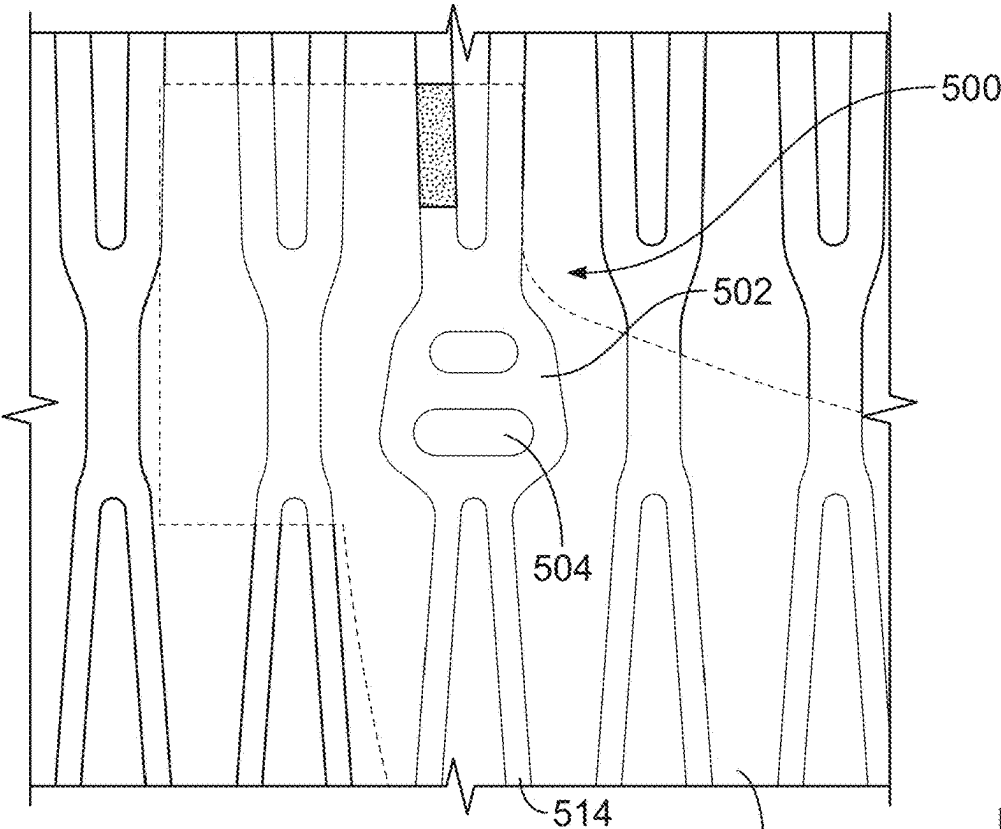


FIG. 7A

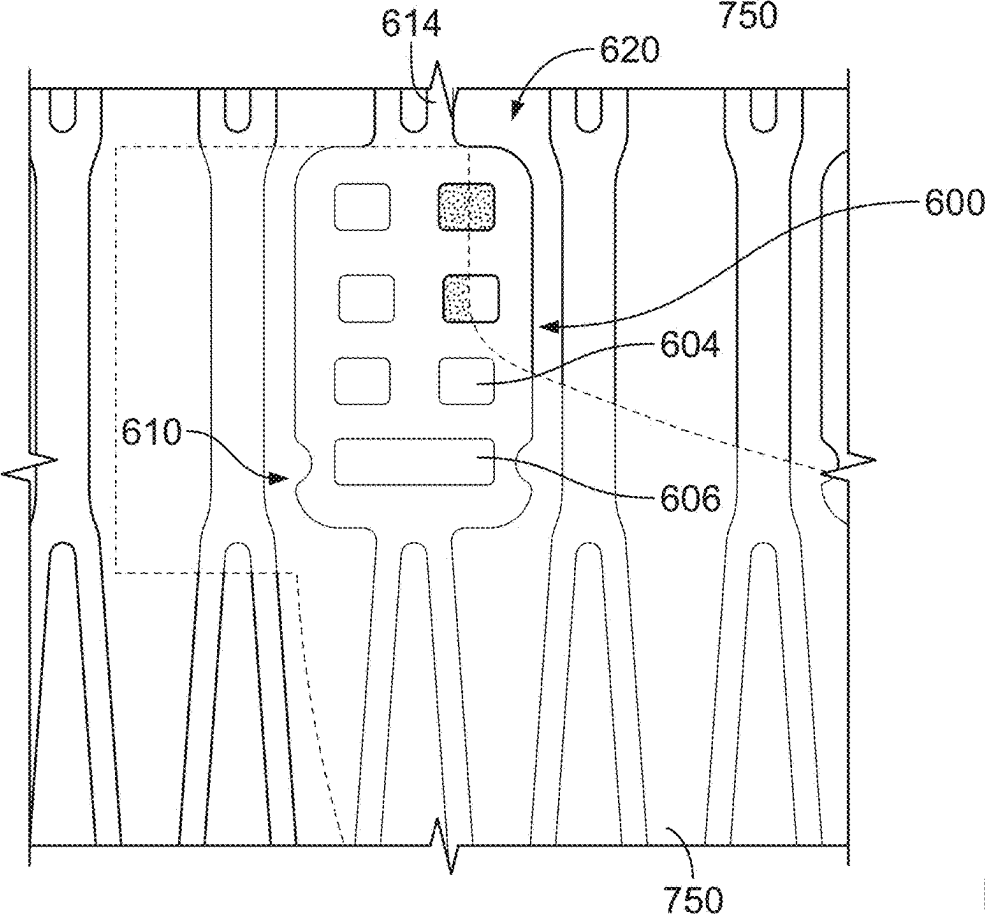


FIG. 7B

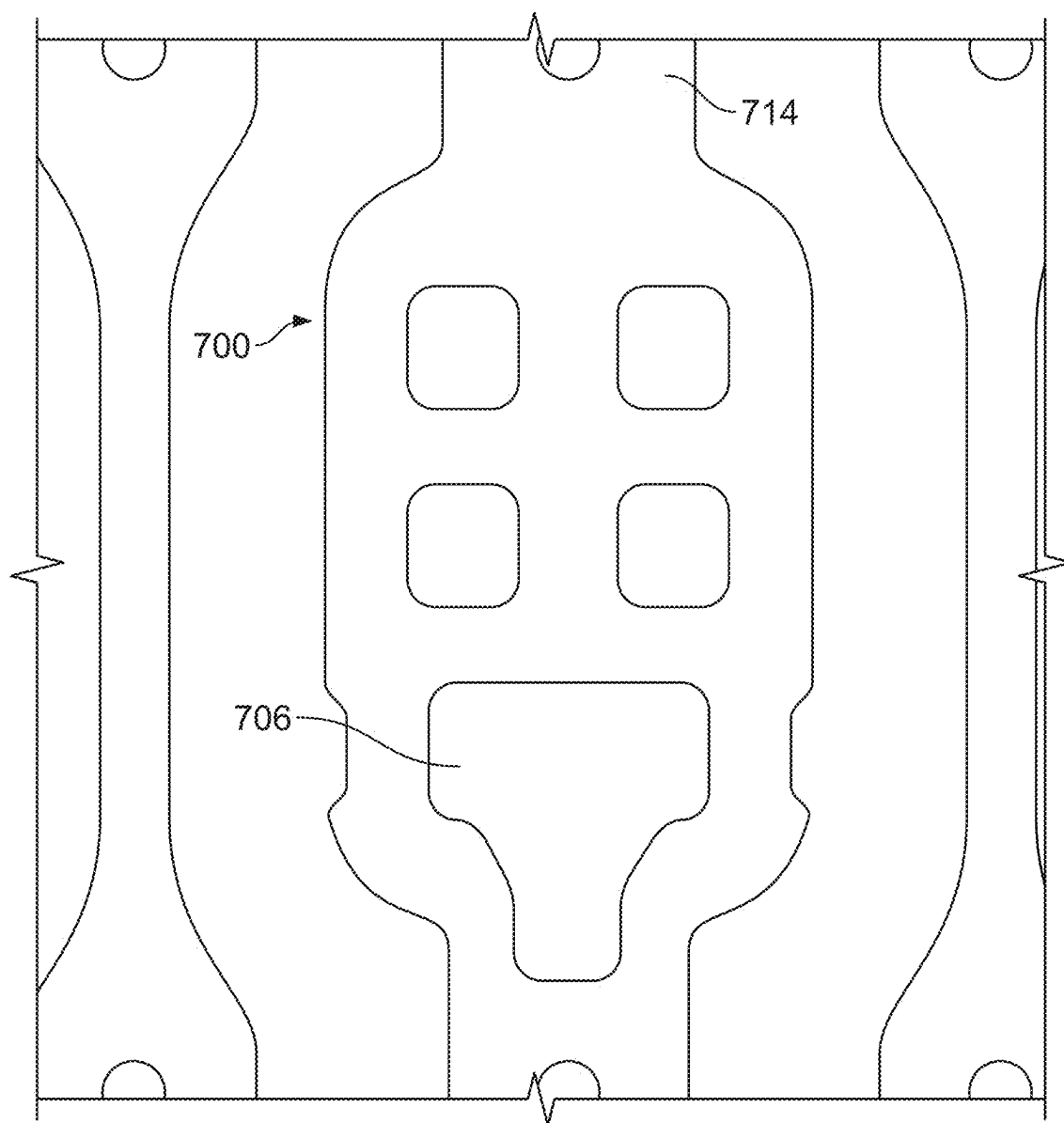
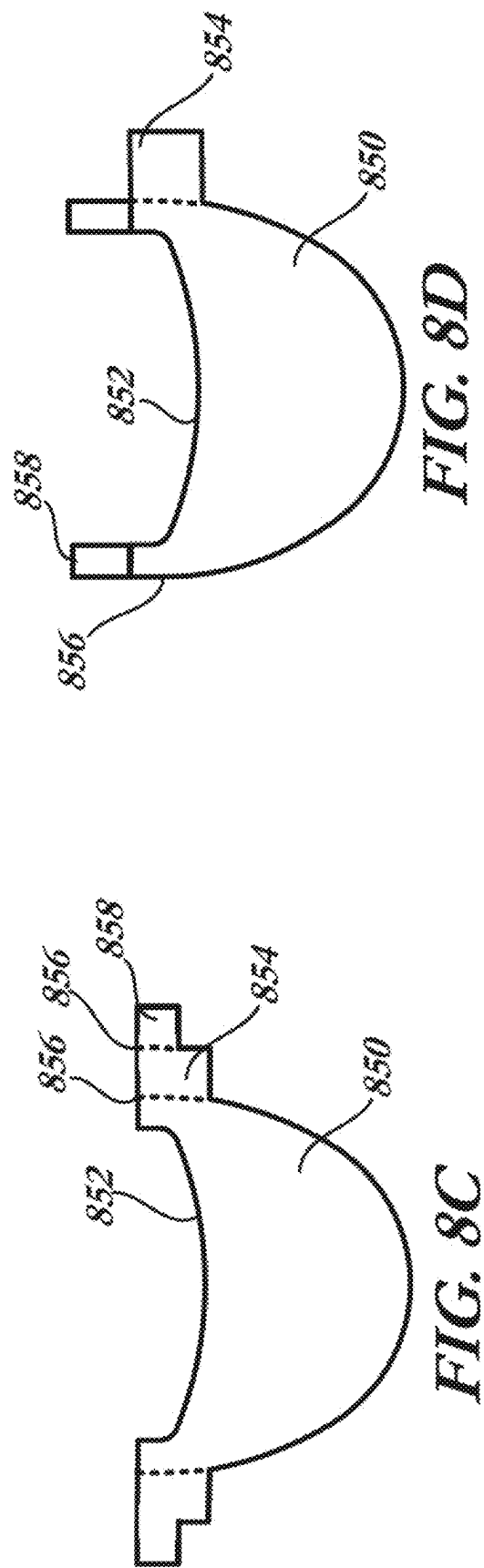
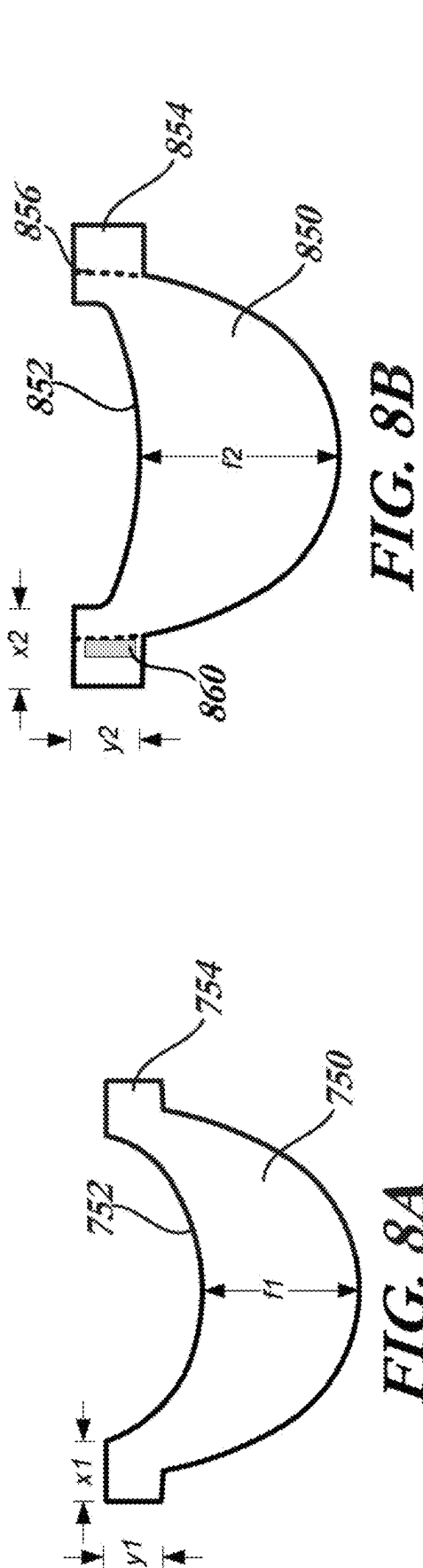


FIG. 7C



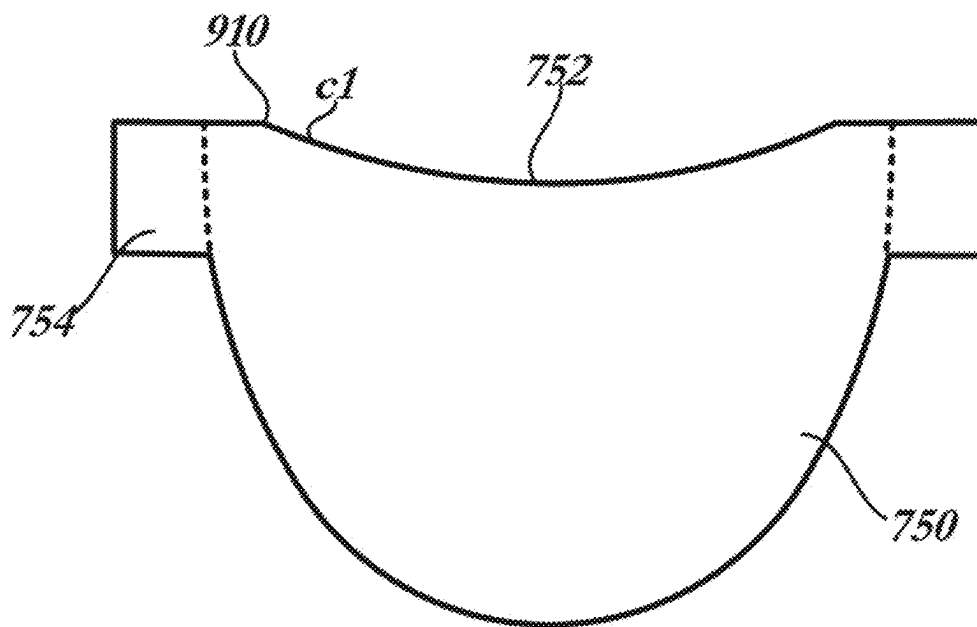


FIG. 9A

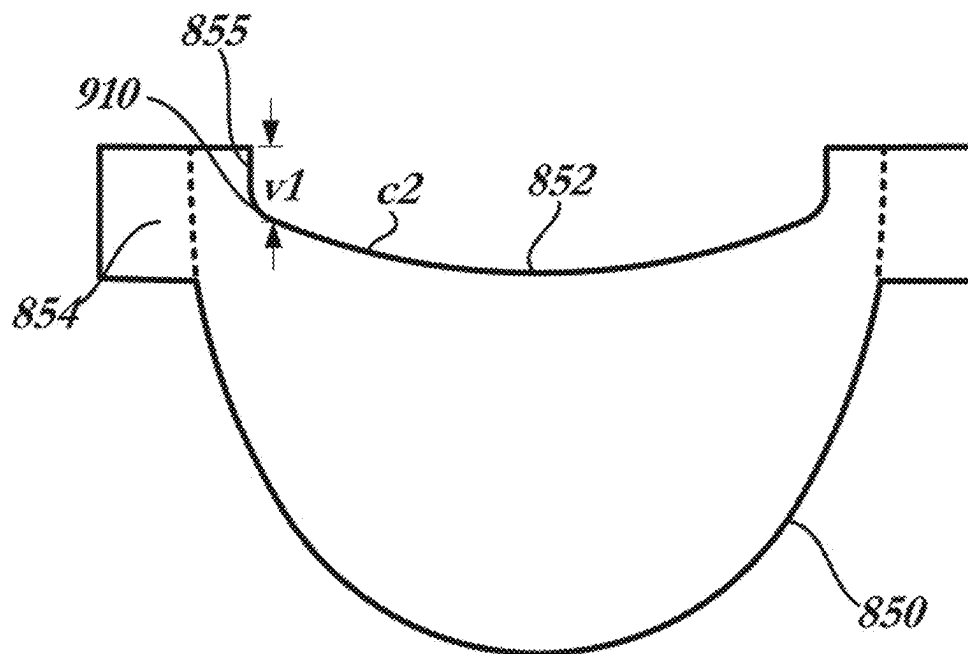


FIG. 9B

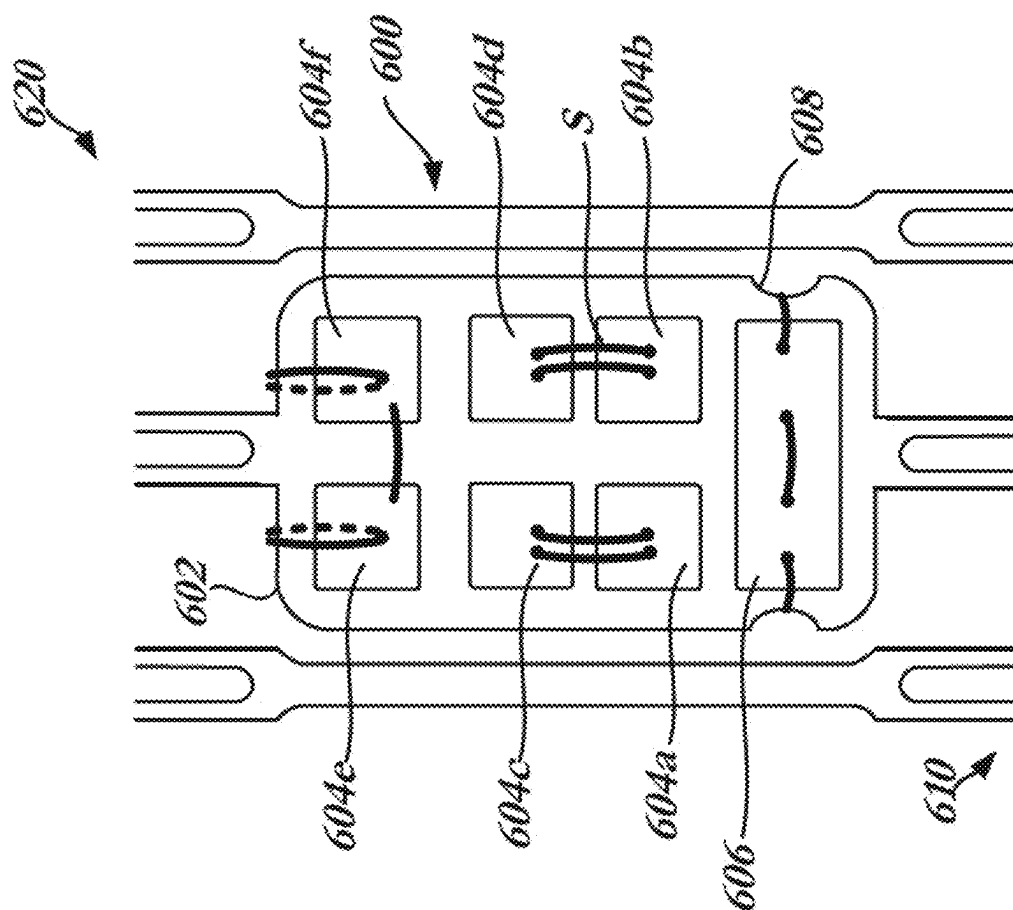


FIG. 10A

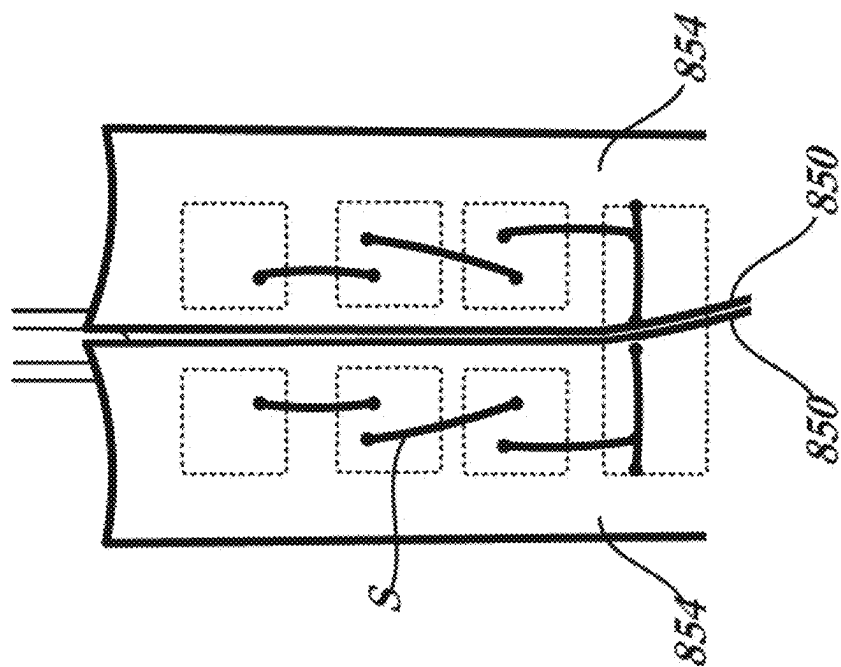


FIG. 10B

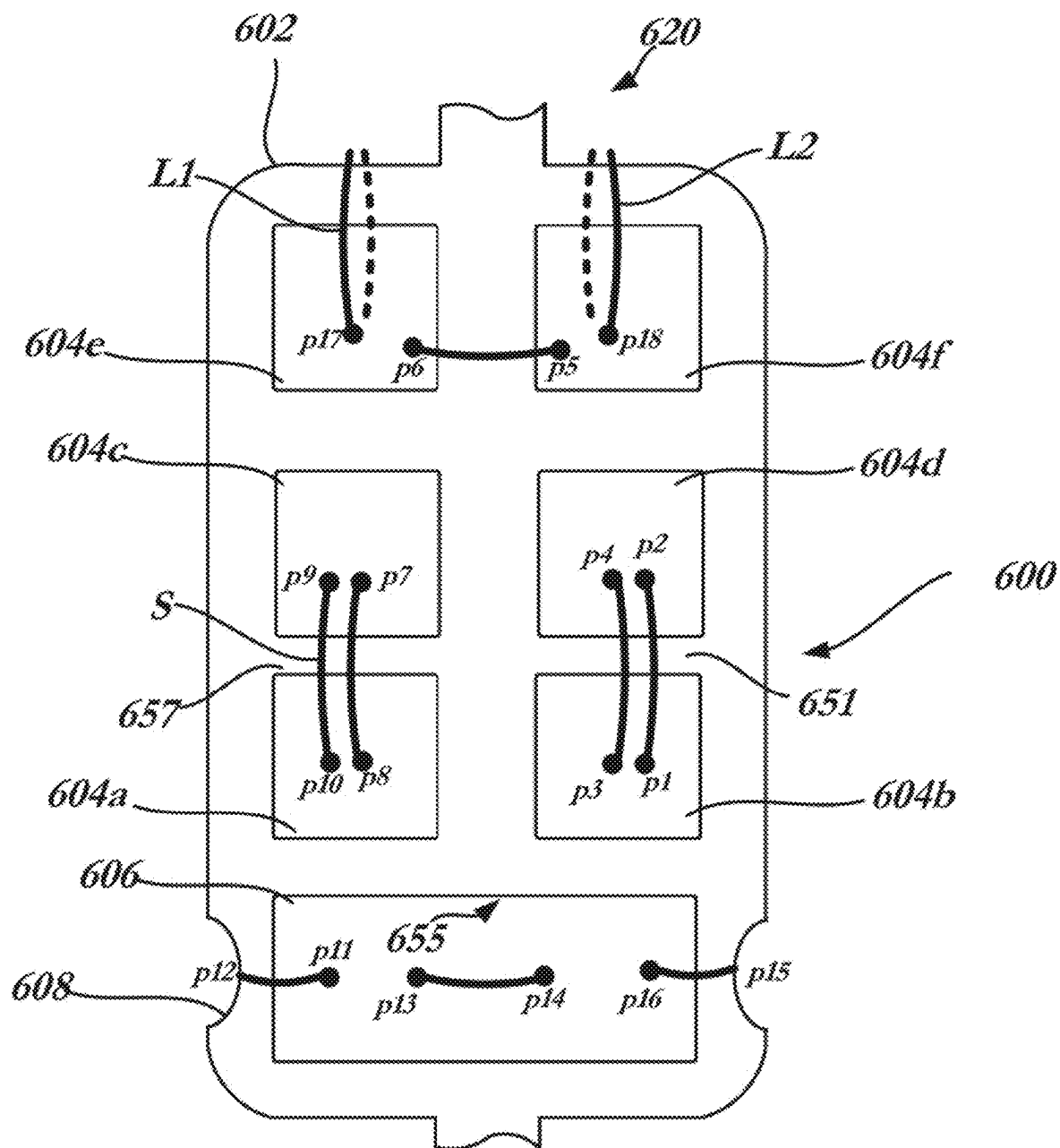


FIG. 10C

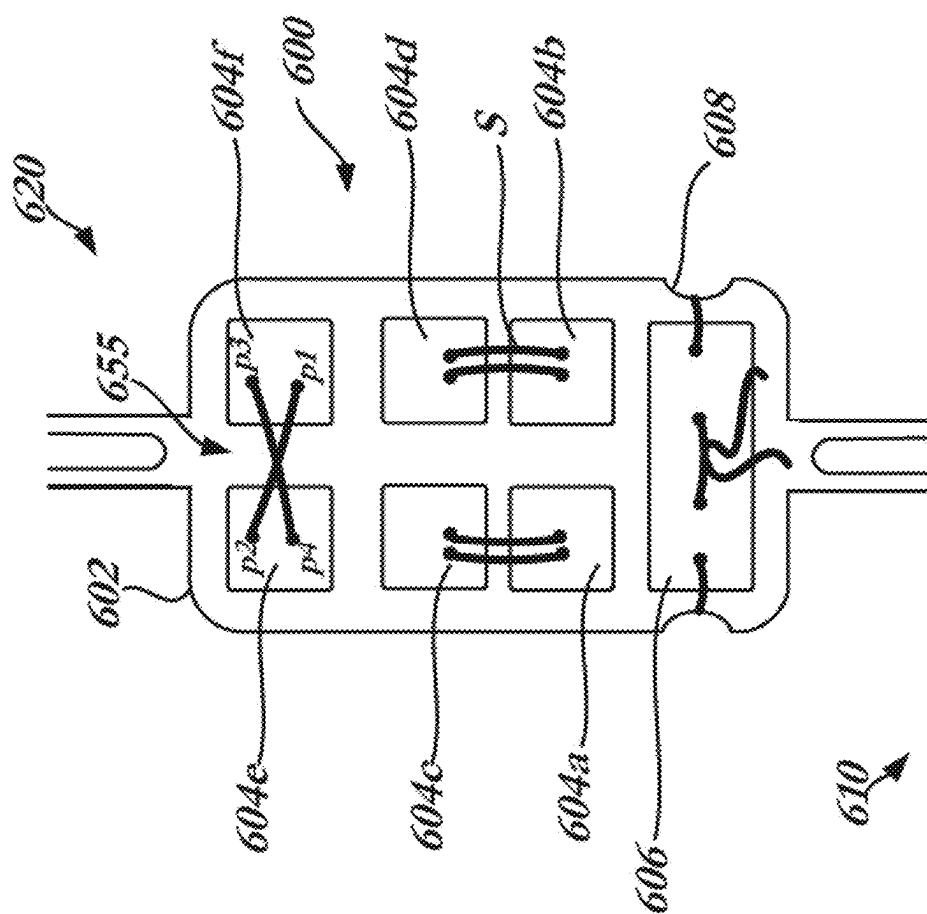


FIG. 11A

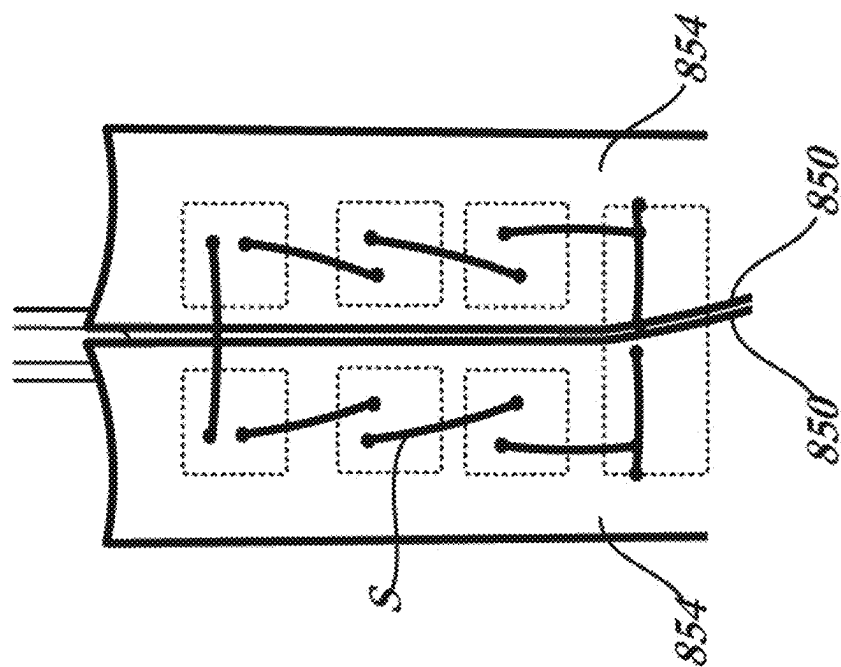


FIG. 11B

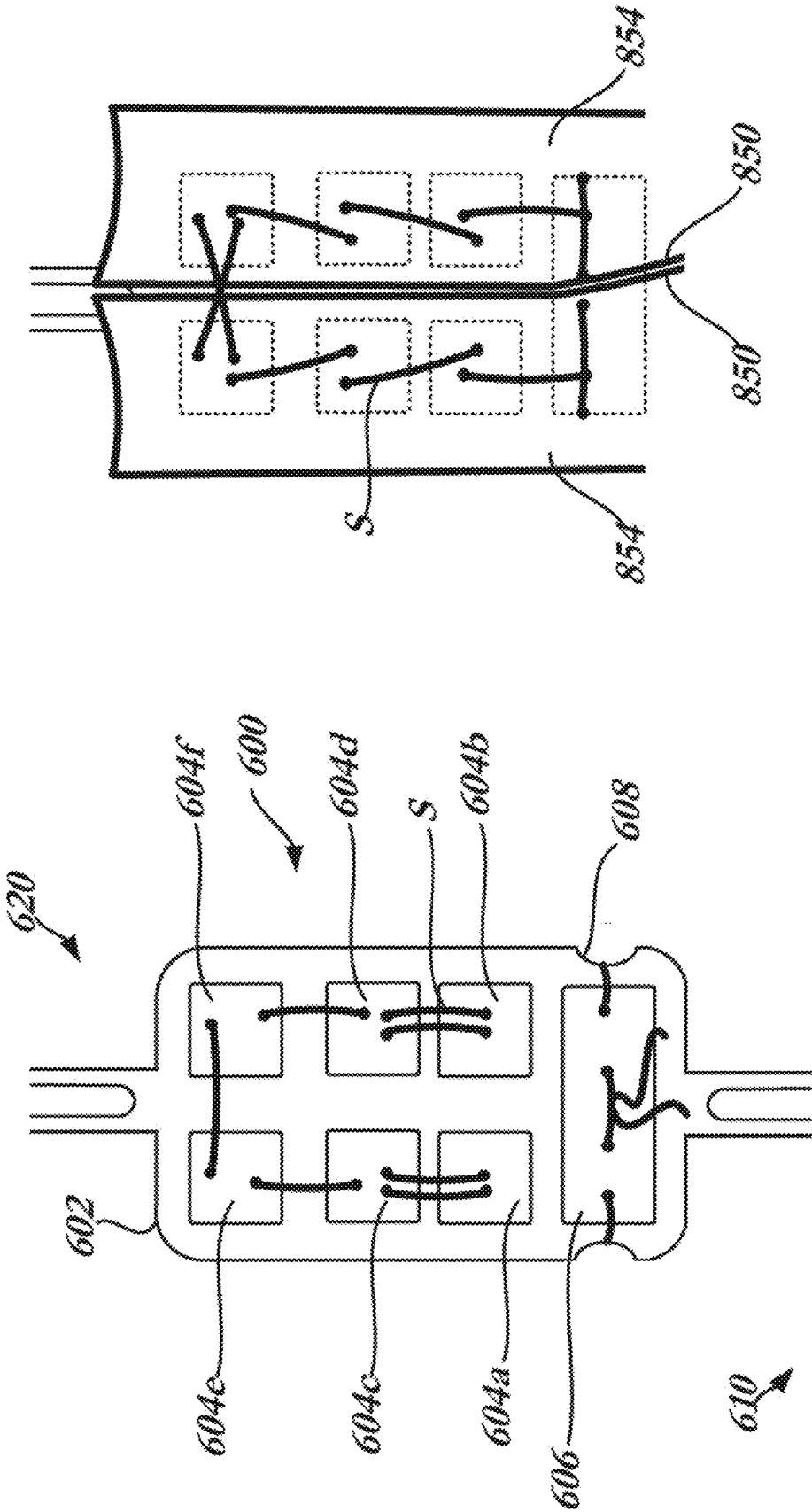


FIG. 12B

FIG. 12A

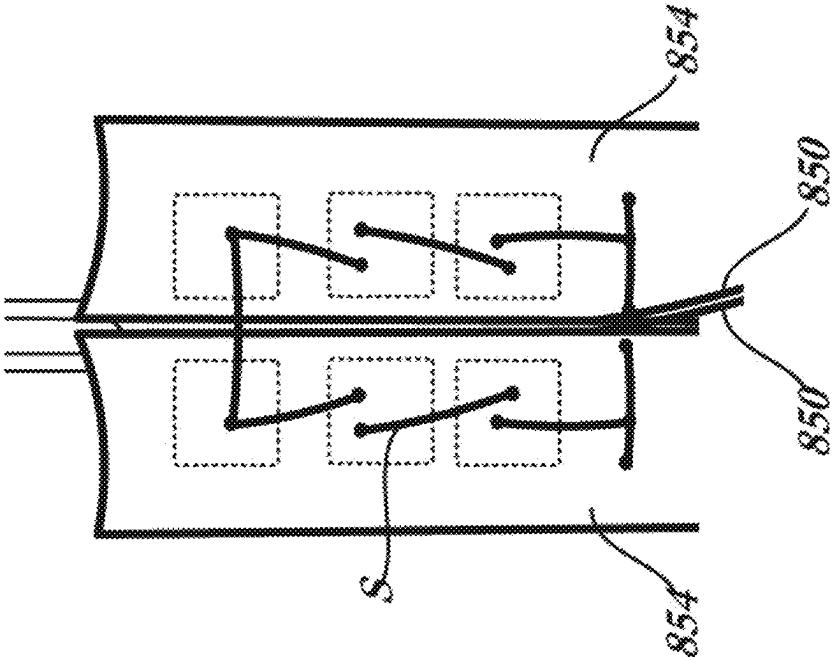


FIG. 13B

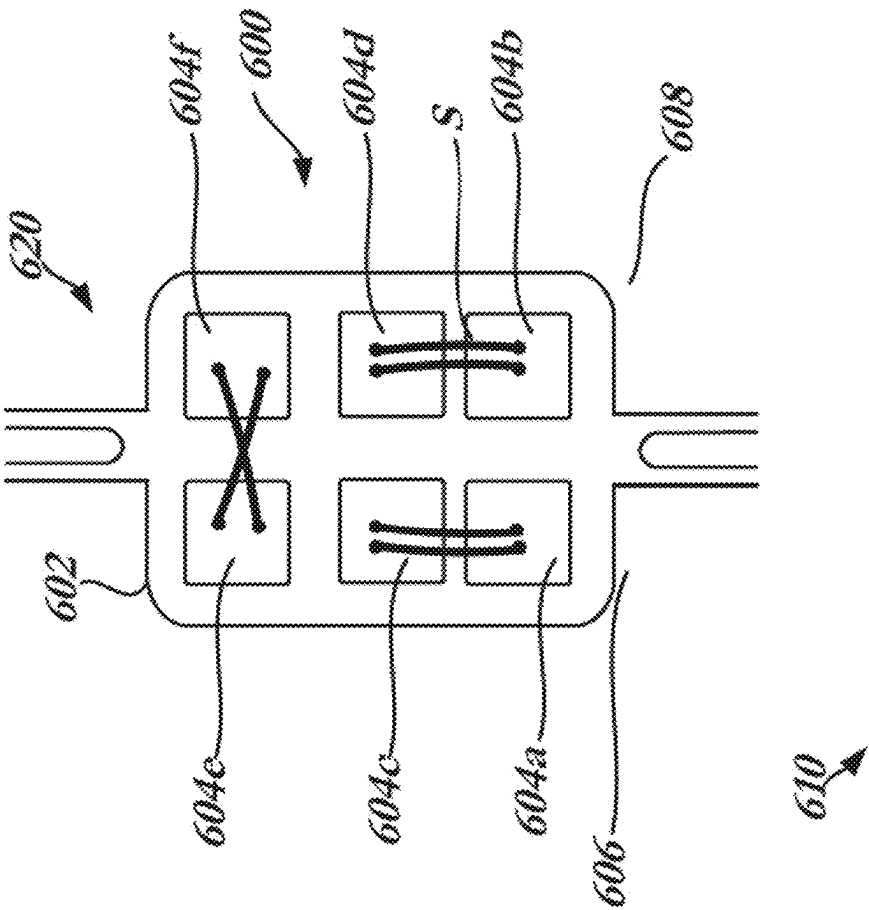


FIG. 13A

COMMISSURE ATTACHMENT FEATURE HAVING MULTIPLE BODIES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a divisional of U.S. patent application Ser. No. 17/871,432 filed Jul. 22, 2022, which is a divisional of U.S. patent application Ser. No. 17/144,248 filed Jan. 8, 2021, which is a divisional of U.S. patent application Ser. No. 15/613,688 filed Jun. 5, 2017, which is a continuation of U.S. patent application Ser. No. 15/202,939 filed Jul. 6, 2016, which is a continuation of U.S. patent application Ser. No. 14/824,551 filed on Aug. 12, 2015, which is a continuation of U.S. patent application Ser. No. 13/781,201 filed on Feb. 28, 2013, which claims the benefit of the filing date of U.S. Provisional Patent Application No. 61/666,224 filed Jun. 29, 2012, the disclosures of which are all hereby incorporated by reference herein.

BACKGROUND OF THE INVENTION

[0002] The present invention relates to heart valve replacement and, in particular, to collapsible prosthetic heart valves. More particularly, the present invention relates to collapsible prosthetic heart valves that may be repositioned during a deployment procedure.

[0003] Prosthetic heart valves that are collapsible to a relatively small circumferential size can be delivered into a patient less invasively than valves that are not collapsible. For example, a collapsible valve may be delivered into a patient via a tube-like delivery apparatus such as a catheter, a trocar, a laparoscopic instrument, or the like. This collapsibility can avoid the need for a more invasive procedure such as full open-chest, open-heart surgery.

[0004] Collapsible prosthetic heart valves typically take the form of a valve structure mounted on a stent. There are two types of stents on which the valve structures are ordinarily mounted: a self-expanding stent and a balloon-expandable stent. To place such valves into a delivery apparatus and ultimately into a patient, the valve must first be collapsed or crimped to reduce its circumferential size.

[0005] When a collapsed prosthetic valve has reached the desired implant site in the patient (e.g., at or near the annulus of the patient's heart valve that is to be replaced by the prosthetic valve), the prosthetic valve can be deployed or released from the delivery apparatus and re-expanded to full operating size. For balloon-expandable valves, this generally involves releasing the entire valve, assuring its proper location, and then expanding a balloon positioned within the valve stent. For self-expanding valves, on the other hand, the stent automatically expands as the sheath covering the valve is withdrawn.

[0006] In delivery systems for self-expanding aortic valves, after the delivery system has been positioned for deployment, the annulus end of the valve may be unsheathed and expanded first, while the aortic end of the valve remains sheathed. Once the annulus end of the valve has expanded, it may be determined that the valve needs to be repositioned in the patient's aortic annulus. To accomplish this, a user (such as a surgeon or an interventional cardiologist) typically resheaths the annulus end of the valve, so that the valve can be repositioned while in a collapsed state. After the valve has been repositioned, the user can fully deploy the valve.

[0007] Once a self-expanding valve has been fully deployed, it expands to a diameter larger than that of the sheath that previously contained the valve in the collapsed condition, making resheathing impossible, or difficult at best. In order for the user to be able to more readily resheath a valve, it is preferable that the position and operation of the valve be evaluated after the valve has been only partially deployed, with a portion of the valve still collapsed inside of the sheath.

[0008] Despite the various improvements that have been made to collapsible prosthetic heart valves, conventional devices, suffer from some shortcomings. For example, in certain procedures, collapsible valves may be implanted in a native valve annulus without first resecting the native valve leaflets. The collapsible valves may have critical clinical issues because of the nature of the stenotic leaflets that are left in place. Additionally, patients with uneven calcification, bi-cuspid disease, and/or valve insufficiency could not be treated well, if at all, with the current collapsible designs.

[0009] The reliance on evenly calcified leaflets could lead to several problems such as: (1) perivalvular leakage (PV leak), (2) valve migration, (3) mitral valve impingement, (4) conduction system disruption, (5) coronary blockage, etc., all of which can have severely adverse clinical outcomes. To reduce these adverse events, the optimal valve would seal and anchor adequately without the need for excessive radial force, protrusion into the left ventricular outflow tract (LVOT), etc., that could harm nearby anatomy and physiology.

[0010] There therefore is a need for further improvements to the devices and systems of collapsible prosthetic heart valves, and in particular, self-expanding prosthetic heart valves. Among other advantages, the present invention may address one or more of these needs.

SUMMARY OF THE INVENTION

[0011] In some embodiments, a prosthetic heart valve includes a collapsible and expandable stent having a proximal end, a distal end, an annulus section adjacent the proximal end and an aortic section adjacent the distal end. The heart valve further includes a plurality of commissure features disposed on the stent and a collapsible and expandable valve assembly, the valve assembly including a plurality of leaflets connected to the plurality of commissure features. Each commissure feature includes a body having a proximal end a distal end, and a plurality of eyelets arranged in at least two rows and at least two columns for distributing load from the plurality of leaflets.

[0012] In some examples, each of the plurality of commissure features includes a body that is substantially rectangular. Additionally, each of the plurality of commissure features may include at least three rows of eyelets disposed on the body. Each of the plurality of commissure features may include two columns of eyelets disposed on the body. Each of the plurality of eyelets may be substantially the same shape and/or size.

[0013] In some embodiments, a prosthetic valve assembly for assembly to a collapsible and expandable stent includes a stent having a plurality of commissure features, each commissure feature including a body having a longitudinal direction extending between a proximal end and a distal end, and a plurality of eyelets arranged in a plurality of columns each extending in the longitudinal direction, the body having

a width in a direction orthogonal to the longitudinal direction. The valve assembly may include a plurality of leaflets, each leaflet having a top free edge for coapting with others of the leaflets and including at least one rectangular tab foldable upon itself along a fold line for attaching to a commissure feature at a leaflet-commissure feature junction, the at least one rectangular tab having a width which is at least as long as the width of the body.

[0014] In some examples, the tab includes a distal end extending beyond the distal end of the body when the tab is aligned for attachment to the commissure feature. A flap may be coupled to one side of the tab to provide an additional buffer at the leaflet-commissure feature junction.

[0015] In some embodiments a prosthetic heart valve includes a collapsible and expandable stent having a proximal end, a distal end, an annulus section adjacent the proximal end and an aortic section adjacent the distal end. A plurality of commissure features may be disposed on the stent, each commissure feature including a body having a longitudinal direction extending between a proximal end and a distal end, and a plurality of eyelets arranged in at least two columns extending in the longitudinal direction and at least two rows extending in a direction orthogonal to the longitudinal direction, for distributing load from the plurality of leaflets. A collapsible and expandable valve assembly may include a plurality of leaflets connected to the plurality of commissure features via a suture pattern. The plurality of eyelets may be arranged in two columns and the suture pattern forms an “X” pattern between an eyelet in a first of the column and an adjacent eyelet of a second in the columns.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] Various embodiments of the present invention are described herein with reference to the drawings, wherein:

[0017] FIG. 1 is a partial side elevational view of a prosthetic heart valve including a valve assembly and a stent;

[0018] FIG. 2A is an end view of a prosthetic heart valve as seen from the aortic sinus toward the heart and the native valve annulus, the valve having in a circular configuration;

[0019] FIG. 2B is an end view of a prosthetic heart valve as seen from the aortic sinus toward the heart and the native valve annulus, the valve having an elliptical configuration;

[0020] FIGS. 3A-C are end views of the prosthetic heart valve of FIG. 1 in various less than ideal configurations due to anatomical differences in the native valve morphology from patient to patient;

[0021] FIG. 4 is a partial side elevational view of the prosthetic heart valve of FIG. 1 depicting areas of high stress on the valve assembly;

[0022] FIG. 5A is an enlarged, schematic side view of a commissure feature of a prosthetic heart valve showing the distribution of load therein;

[0023] FIG. 5B is an enlarged, schematic side view of a commissure feature showing the load distribution therein according to one embodiment of the present invention;

[0024] FIGS. 6A-S are enlarged schematic side views of several embodiments of commissure features according to the present invention;

[0025] FIG. 7A is an enlarged, schematic side view of the commissure feature of FIG. 5A being attached to a leaflet;

[0026] FIG. 7B is an enlarged, schematic side view of the commissure feature of FIG. 6F being attached to a leaflet;

[0027] FIG. 7C is an enlarged, schematic side view of another example of a commissure feature according to the present invention;

[0028] FIG. 8A is an enlarged, schematic side view of a conventional valve leaflet;

[0029] FIG. 8B is an enlarged, schematic side view of a first embodiment of a valve leaflet according to the present invention;

[0030] FIG. 8C is an enlarged, schematic side view of a second embodiment of a valve leaflet according to the present invention;

[0031] FIG. 8D is an enlarged, schematic side view of a third embodiment of a valve leaflet according to the present invention;

[0032] FIGS. 9A-B are enlarged, schematic side views of a conventional leaflet and a leaflet according to the present invention, showing the tab-free edge attachment;

[0033] FIGS. 10A-B are enlarged, schematic side views showing a suture pattern attaching leaflets to a commissure feature as shown from the abluminal side and the luminal side of the valve, respectively;

[0034] FIG. 10C is an enlarged, schematic view of FIG. 10A showing the attachment points of the suture pattern in more detail;

[0035] FIGS. 11A-B are enlarged, schematic side views showing another suture pattern attaching leaflets to a commissure feature as shown from the abluminal side and the luminal side of the valve respectively;

[0036] FIGS. 12A-B are enlarged, schematic side views showing a further suture pattern attaching leaflets to a commissure feature as shown from the abluminal side and the luminal side of the valve respectively; and

[0037] FIGS. 13A-B are enlarged, schematic side views showing yet another suture pattern attaching leaflets to a commissure feature as shown from the abluminal side and the luminal side of the valve respectively.

[0038] Various embodiments of the present invention will now be described with reference to the appended drawings. It is to be appreciated that these drawings depict only some embodiments of the invention and are therefore not to be considered limiting of its scope.

DETAILED DESCRIPTION OF THE INVENTION

[0039] As used herein, the term “proximal,” when used in connection with a prosthetic heart valve, refers to the end of the heart valve closest to the heart when the heart valve is implanted in a patient, whereas the term “distal,” when used in connection with a prosthetic heart valve, refers to the end of the heart valve farthest from the heart when the heart valve is implanted in a patient.

[0040] FIG. 1 shows a collapsible prosthetic heart valve 100 according to an embodiment of the present disclosure. The prosthetic heart valve 100 is designed to replace the function of a native aortic valve of a patient. Examples of collapsible prosthetic heart valves are described in International Patent Application Publication No. WO/2009/042196; U.S. Pat. Nos. 7,018,406; and 7,329,278, the disclosures of all of which are hereby incorporated herein by reference. As discussed in detail below, the prosthetic heart valve has an expanded condition and a collapsed condition. Although the invention is described herein as applied to a prosthetic heart valve for replacing a native aortic valve, the invention is not

so limited, and may be applied to prosthetic valves for replacing other types of cardiac valves.

[0041] The prosthetic heart valve **100** includes a stent or frame **102**, which may be wholly or partly formed of any biocompatible material, such as metals, synthetic polymers, or biopolymers capable of functioning as a stent. Suitable biopolymers include, but are not limited to, elastin, and mixtures or composites thereof. Suitable metals include, but are not limited to, cobalt, titanium, nickel, chromium, stainless steel, and alloys thereof, including nitinol. Suitable synthetic polymers for use as a stent include, but are not limited to, thermoplastics, such as polyolefins, polyesters, polyamides, polysulfones, acrylics, polyacrylonitriles, polyetheretherketone (PEEK), and polyaramides. The stent **102** may have an annulus section **110** an aortic section (not shown) and a transition section (not shown) disposed between the annulus section and the aortic section. Each of the annulus section **110**, the transition section and the aortic section of the stent **102** includes a plurality of cells **112** connected to one another around the stent. The annulus section **110** and the aortic section of the stent **102** may include one or more annular rows of cells **112** connected to one another. For instance, the annulus section **110** may have two annular rows of cells **112**. When the prosthetic heart valve **100** is in the expanded condition, each cell **112** may be substantially diamond shaped. Regardless of its shape, each cell **112** is formed by a plurality of struts **114**. For example, a cell **112** may be formed by four struts **114**.

[0042] The stent **102** may include commissure features **116** connecting at least two cells **112** in the longitudinal direction of the stent **102**. The commissure features **116** may include eyelets for facilitating the suturing of a valve assembly **104** to the stent **102**.

[0043] The prosthetic heart valve **100** also includes a valve assembly **104** attached inside the annulus section **110** of the stent **102**. United States Patent Application Publication No. 2008/0228264, filed Mar. 12, 2007, and United States Patent Application Publication No. 2008/0147179, filed Dec. 19, 2007, the entire disclosures of both of which are hereby incorporated herein by reference, describe suitable valve assemblies. The valve assembly **104** may be wholly or partly formed of any suitable biological material or polymer. Examples of biological materials suitable for the valve assembly **104** include, but are not limited to, porcine or bovine pericardial tissue. Examples of polymers suitable for the valve assembly **104** include, but are not limited to, polyurethane and polyester. In at least some examples, portions of valve assembly **104**, a cuff and the suture used may include an ultra high molecular weight polyethylene, such as FORCE FIBER®.

[0044] The valve assembly **104** may include a cuff **106** disposed on the luminal surface of annulus section **110**, on the abluminal surface of annulus section **110**, or on both surfaces, and the cuff may cover all or part of either or both of the luminal and abluminal surfaces of the annulus section. The cuff **106** and/or the sutures used to attach the valve assembly **104** to stent **102** may be formed from or include ultra-high-molecular-weight polyethylene. FIG. 1 shows cuff **106** disposed on the luminal surface of annulus section **110** so as to cover part of the annulus section while leaving another part thereof uncovered. The cuff **106** may be attached to stent **102** by one or more strings or sutures passing through the cuff and around selected struts **114** of the stent. The valve assembly **104** may further include a plu-

rality of leaflets **108** which collectively function as a one-way valve. A first edge **122** of each leaflet **108** may be attached to the stent **102** between two adjacent commissure features **116** by any suitable attachment means, such as suturing, stapling, adhesives, laser, heat or ultrasonic bonding or the like. For example, the first edge **122** of each leaflet **108** may be sutured to the stent **102** by passing strings or sutures through the cuff **106** of the valve assembly **104**. The leaflets **108** may be attached to the stent **102** along at least some struts **114** of the stent and through the eyelets in the commissure features **116** to enhance the structural integrity of the valve assembly **104**. A second or free edge **124** of each leaflet **108** may coapt with the corresponding free edges of the other leaflets, thereby enabling the leaflets to function collectively as a one-way valve.

[0045] As shown in FIG. 1, at least one leaflet **108** may be attached to the stent **102** so that its first edge **122** is disposed substantially along specific struts **114a**, **114b**, **114c**, **114d**, **114e** and **114f**/located in the annulus section **110** of the stent. That is, the edge **122** is positioned in substantial alignment with struts **114a**, **114b**, **114c**, **114d**, **114e**, and **114f**. Struts **114a**, **114b**, and **114c** may be connected to one another in substantially end-to-end fashion diagonally along three cells **112**, beginning with an end of the strut **114a** connected to a commissure feature **116** and ending with an end of strut **114c** connected to an end of strut **114d**. Struts **114c** and **114d** are part of the same cell **112** and may collectively define a substantially right angle between them. Struts **114d**, **114e**, and **114f** may be connected to one another in substantially end-to-end fashion diagonally along three cells **112**, beginning with an end of the strut **114f** connected to a commissure feature **116** and ending with the connection between an end of strut **114c** and an end of strut **114d**.

[0046] As discussed above, the leaflets **108** may be attached directly to and supported by the struts **114a**, **114b**, **114c**, **114d**, **114e**, and **114f**, and by commissure features **116**, such as by suturing. In such event, the cuff **106** may perform little or no supportive function for the leaflets **108**. Hence, the cuff **106** is not subjected to high stresses and is therefore less likely to wear during use. In light of this, the thickness of the cuff may be reduced. Reducing the thickness of the cuff **106** results in a decrease in the volume of the valve assembly **104** in the collapsed condition. This decreased volume is desirable as it enables the prosthetic heart valve **100** to be implanted in a patient using a delivery device that is smaller in cross-section than conventional delivery devices. In addition, since the material forming the stent struts **114** is stronger than the material forming the cuff **106**, the stent struts **114** may perform the supportive function for the leaflets **108** better than the cuff **106**.

[0047] The volume of the valve assembly **104** may be further reduced by having the cuff **106** cover only a portion of the surface of annulus section **110**. With continued reference to FIG. 1, the first or proximal end **118** of the cuff **106** may substantially follow the contour of the first or proximal end **119** of the stent **102**. As such, the proximal end of the cuff **106** may have a generally sinusoidal or zigzag shape. This eliminates any free edge of the cuff **106**, which otherwise might extend directly between the cusps of the cells **112** at the proximal end **119** of the stent **102**, and enables the entire length of the proximal end **118** of the cuff **106** to be secured to the stent **102**. The second or distal end **120** of the cuff **106**, on the other hand, may be disposed substantially along at least some struts **114**, but not neces-

sarily the struts in a single annular row of cells **112**. More particularly, the distal end **120** of the cuff **106** may follow the stent struts **114** up to the commissure features **116**, such that the cuff covers all of the cells **112** in the bottom annular row **113** of cells and in a second annular row **115** of cells located between the commissure features and the proximal end **119** of the stent **102**, but covers a lesser area of cells in the annular regions between the commissure features. In other words, the distal end **120** of the cuff **106** may be disposed substantially along struts **114a**, **114b**, **114e**, **114f**, **114g** and **114h**, as shown in FIG. 1. Strut **114g** may be connected at one end to strut **114h**, and at the other end to the intersection of struts **114b** and **114c**. Strut **114h** may be connected at one end to strut **114g**, and at the other end to the intersection of struts **114d** and **114e**. Struts **114c**, **114d**, **114g**, and **114h** collectively form a single cell **112**.

[0048] As a result of the foregoing configuration, all of the cells **112** in the bottom annular row **113** of cells may be entirely covered by the cuff **106**. The cuff **106** may also entirely cover those cells **112** in the second annular row **115** that are located directly below the commissure features **116**. All of the other cells **112** in the stent **102** may be open or not covered by the cuff **106**. Hence, there may be no cells **112** which are only partially covered by the cuff **106**.

[0049] Since the edges of the valve leaflets **108** extend up to the second annular row **115** of cells **112** only in the regions of the commissure features **116**, there is little to no likelihood of leakage in the area of the cells between the commissure features in the second annular row of cells, and therefore no need for the cuff **106** to cover this area. This reduction in the area of the cuff **106**, both at the proximal end **118** and at the distal end **120** thereof, reduces the amount of material in the valve assembly **104**, thereby enabling the prosthetic valve **100** to achieve a smaller cross-section in the collapsed condition.

[0050] In operation, the embodiments of the prosthetic heart valve **100** described above may be used to replace a native heart valve, such as the aortic valve, a surgical heart valve or a heart valve that has undergone a surgical procedure. The prosthetic heart valve may be delivered to the desired site (e.g., near a native aortic annulus) using any suitable delivery device. During delivery, the prosthetic heart valve is disposed inside the delivery device in the collapsed condition. The delivery device may be introduced into a patient using a transfemoral, transapical, transseptal, tranxaxillary or other approach. Once the delivery device has reached the target site, the user may deploy the prosthetic heart valve. Upon deployment, the prosthetic heart valve expands into secure engagement within the native aortic annulus. When the prosthetic heart valve is properly positioned inside the heart, it works as a one-way valve, allowing blood to flow in one direction and preventing blood from flowing in the opposite direction.

[0051] In certain procedures, collapsible valves may be implanted in a native valve annulus without first resecting the native valve leaflets. The collapsible valves may have critical clinical issues because of the nature of the stenotic leaflets that are left in place. Additionally, patients with uneven calcification, bi-cuspid aortic valve disease, and/or valve insufficiency could not be treated well, if at all, with the current collapsible designs.

[0052] The reliance on evenly calcified leaflets could lead to several problems such as: (1) perivalvular leakage (PV leak), (2) valve migration, (3) mitral valve impingement, (4)

conduction system disruption, (5) coronary blockage, etc., all of which can have severely adverse clinical outcomes. To reduce these adverse events, the optimal valve would seal and anchor adequately without the need for excessive radial force, protrusion into the left ventricular outflow tract (LVOT), etc., that could harm nearby anatomy and physiology.

[0053] FIGS. 2A and 2B show an end view of a prosthetic heart valve **200** as seen from the outflow or downstream side of the valve assembly, e.g., looking from the aorta or aortic sinus toward the heart and the native valve annulus. As seen in FIG. 2A, the valve assembly includes a stent **202** and valve leaflets **208a**, **208b**, and **208c** attached to the stent and to commissure features **216a**, **216b**, and **216c**. At least one edge of each leaflet **208** is sutured to the stent **202** and to two of the three commissure features **216**, leaving at least one edge free to move in response to the pumping of blood. As the blood pressure in the left ventricle increases, the free edges of the leaflets move away from one another to allow blood to flow from the left ventricle to the aorta, following which the free edges move toward one another and coapt to prevent blood from flowing back from the aorta into the left ventricle.

[0054] It will be understood that the coaptation of “the free edges” of the valve leaflets does not necessarily mean that the actual edges meet per se. Indeed, the leaflets are preferably sized, shaped, and attached such that a suitable “belly” contour is formed. And the leaflets should each include a portion extending from the free edge toward the annulus (referred to herein as a “coaptation section”) that may engage the coaptation sections of the other leaflets such that there will be a surface area of contact between the leaflets rather than edge-to-edge contact. This surface area of contact is important so that, when in a closed or “coapted” condition, the leaflets cooperate to substantially prevent backflow or regurgitation of blood through the valve. These areas of actual contact between the coaptation sections of adjacent leaflets are referred to herein as the coaptation junctions of the leaflets and are illustrated in FIG. 2A at **211a**, **211b**, and **211c**. The coaptation section of each leaflet may range in size as a particular valve design demands, but generally will be sufficient to provide some tolerance or ability to form a coaptation junction even if the shape of the valve is distorted during placement, as illustrated in FIG. 2B.

[0055] The annulus section of prosthetic heart valve **200** has a generally regular cylindrical shape by which is meant that the structure has a generally circular cross-section with a substantially constant diameter along its length. When placed in the annulus of a native heart valve, such as, for example, the tricuspid aortic valve, and expanded, a substantially fluid-tight fit should result. However, the native valve annulus may not be circular, and, in fact, may vary from patient to patient, as may the shape of the aortic sinus or aorta, the angle of the junction between the valve annulus and the aortic sinus, and other local anatomical features. When prosthetic heart valve **200** is deployed and expanded, it must accommodate these anatomical variations in order to function properly. This may result in a distortion in the shape of stent **202** and/or valve assembly **204**, and the repositioning of leaflets **208a**, **208b**, and **208c** relative to one another, which can affect the coaptation junctions **211a**, **211b**, and **211c**.

[0056] As the stent of a collapsible prosthetic heart valve distorts during implantation, during beating of the heart, or

because of irregularities in the patient's anatomy or the condition of the native valve, such distortion may be translated to the valve assembly, such that not all of the valve leaflets meet to form effective coaptation junctions. This can result in leakage or regurgitation and other inefficiencies which can reduce cardiac performance. Moreover, if the prosthetic valve is not placed optimally and the valve leaflets are not coapting as intended, other long term effects, such as uneven wear of the individual leaflets, can be postulated.

[0057] As shown in FIG. 2B, ideally, valve leaflets **208a**, **208b**, and **208c** fully coapt despite the distortion of the annulus section (hidden behind the valve leaflets in this figure) into a more elongated or elliptical configuration. As will be appreciated, the distortion of the annulus section affects the relative positions of commissure features **216a-c**, as well as the positions of leaflets **208a-c** relative to one another. The ability of the valve leaflets **208a-c** to fully coapt despite this distortion enables prosthetic valve **200** to function in the manner intended.

[0058] FIGS. 3A-C illustrate a series of prosthetic heart valve as seen from the downstream side of the valve assembly, e.g., looking from the aorta or aortic sinus toward the heart and the native valve annulus. Each of these valves has an elliptical configuration resulting from the shape of the native valve annulus, calcification and possibly other defects. As seen in FIGS. 3A-C, the imperfect geometry the valve assumes results in inadequate coaptation of the leaflets **208a-c**. Specifically, the leaflets **208** of these valves are incapable of complete coaptation when disposed in a native valve annulus with an elliptical, ovoid or otherwise non-circular configuration. In some examples, a gap **238** may be evident between the leaflets **208** in what should be the fully closed condition. Such inadequate coaptation may lead to leakage and regurgitation as discussed above.

[0059] In addition to inadequate coaptation, less than ideal native valve geometry may also increase the stresses on certain portions of the prosthetic heart valve. FIG. 4 is a partial side view of a prosthetic heart valve **400** having a stent **402** and a valve assembly **404** disposed in the annulus section **410** of the stent. Within the heart valve **400**, leaflets **408** are attached to cuff **406** via sutures. Specifically, FIG. 4 shows the load distribution in the valve assembly. When leaflets **408** coapt to form a closed configuration, load is transferred from the leaflet structure to the leaflet-commissure feature junction as indicated by "A". The load distribution diagram shows that high point loads are generated at region "A" where the leaflets are joined to the commissure feature **416**. If the point loads at regions "A" are sufficiently high, the leaflets may tear from the commissure feature. Thus, regions A may be prone to failure.

[0060] Features of this aspect of the present invention will be described in connection with the commissure features shown in FIGS. 5A and 5B. It will also be noted that while the inventions herein described are predominately discussed in terms of a tricuspid valve and a stent having a shape as illustrated in FIG. 1, the valve could be a bicuspid valve, such as the mitral valve, and the stent could have different shapes, such as a flared or conical annulus section, a less-bulbous aortic section, and the like, and a differently shaped transition section.

[0061] FIG. 5A illustrates a previous version of a commissure feature **500** coupled to struts **514** for attaching a valve assembly to the stent. Commissure feature **500** is formed of a body **502** having a pair of eyelets **504**. Leaflets

(not shown) may be attached via sutures to commissure feature **500** through eyelets **504** and struts **514**. Load is distributed across the area of commissure feature **500** as shown in FIG. 5A. Specifically, stress from the leaflets is distributed across area **L1** of the commissure feature **500**.

[0062] FIG. 5B illustrates a commissure feature **600** coupled to struts **614** according to one embodiment of the present invention. Commissure feature **600** is formed of a body **602** having a plurality of eyelets **604** arranged in rows and/or columns. Leaflets (not shown) may be attached via sutures, glue, staples or any suitable means to commissure feature **600** through eyelets **604** and struts **614**. Stress from the leaflets is distributed across area **L2** of the commissure feature **600**, as shown in FIG. 5B. Details of commissure feature **600** will be described in greater detail with reference to FIGS. 6A-J. It will suffice to note at this point that a comparison of FIGS. 5A and 5B illustrates the difference in the area for load distribution between commissure features **500** and **600**. Specifically, commissure feature **600** provides a larger area **L2** across which the stress from the leaflets is distributed when compared to area **L1** of commissure features **500**. A larger area for distributing loads may decrease the chance of failure at the commissure feature-leaflet attachment. Moreover, increasing the spacing between eyelets **604** may allow the valve to function better acutely and chronically. The benefit of wider-spaced eyelets may be recognized in improved ease of manufacturing, which in turn generally leads to improved yields for functional and visual valve criteria. Ease of manufacturing reduces the likelihood for tolerance stack-ups related to tissue folding and alignment. This may also reduce need for pinching or manipulation of the valve during assembly.

[0063] FIGS. 6A-J illustrates several specific embodiments of commissure features according to the present invention, all of which have the same general shape and characteristics. It will be appreciated that the commissure features described in these figures are exemplary and should not be considered limiting. Moreover, the described features in the following embodiments may be combined or modified in any desirable manner.

[0064] FIG. 6A illustrates a commissure feature **600** according to one embodiment of the present invention. Commissure feature **600** includes a body **602** having a proximal end **610** and a distal end **620**, and a plurality of eyelets **604** disposed therein. Specifically, the commissure feature **600** of FIG. 6A includes four rows of eyelets **604** along the length of body **602**, each row including two eyelets so as to form two columns of eyelets. The rows of eyelets **604** may be evenly spaced and of the same shape and size, as illustrated in FIG. 6A, so as to be symmetrical with respect to a central longitudinal axis of body **602**. As depicted, the eyelets **604** are all in the shape of similarly sized squares. Body **602** is coupled to struts **614** at its proximal end **610** and distal end **620**. Leaflets (not shown) may be attached via sutures to commissure feature **600** through eyelets **604**.

[0065] FIG. 6B illustrates a second embodiment of commissure feature **600** which is similar to that of FIG. 6A, with two exceptions. First, recesses **608** are disposed on the side edges of the body **602** near the distal end **620**. Recesses **608** may be formed as indentations or depressions in body **602** and sutures may be wrapped around or disposed within the recesses. Such recesses may be useful in not only securing and guiding a suture, but also in protecting the suture from

adjacent cells, a delivery system or other anatomical bodies that may damage it. Second, the plurality of eyelets **604** are not evenly spaced in FIG. 6B. Though the four rows of eyelets **604** are evenly spaced in the longitudinal direction, each of the first three rows of eyelets (beginning at the proximal end) **610** of body **402** has a pair of eyelets spaced a first distance from the central axis of the body, while the fourth row has a pair of eyelets spaced closer to one another and closer to the central axis of the body. Nonetheless, the pattern of eyelets **604** in this commissure feature **600** is also symmetrical with respect to the central longitudinal axis of body **602**, the row in which the eyelets are spaced closer to one another need not be the fourth row, but may be any row. In addition, the eyelets may be spaced closer to one another in more than one row. Spacing may depend on leaflet thickness and where the free edge transitions so as to not pinch the leaflets too tightly together or abrade the free edge of the leaflet as it moves.

[0066] Another embodiment of commissure feature **600** is depicted in FIG. 6C. In this commissure feature, eyelets **604** are not all the same size. Instead, the first row of eyelets (beginning at the proximal end **610** of body **602**) includes larger square eyelets **604a** and **604b**, while the next three rows include smaller, rectangular eyelets **604c-h**. While depicted in the first row, the larger square eyelets can be in any row, and in more than one row. All four rows of eyelets may be evenly spaced in the longitudinal direction, and the pattern of eyelets is symmetrical with respect to the central longitudinal axis of body **602**.

[0067] FIG. 6D illustrates another embodiment of commissure feature **600** including four rows of eyelets **604** having the same shape but different sizes. The first three rows of eyelets **604** (beginning at the proximal end **610** of body **602**) are all squares of substantially the same size, while the fourth row has square eyelets **604** of a smaller size. The row of eyelets having a different size (smaller or larger) need not be the fourth row, but may be any row. Moreover, the different-sized eyelets need not be limited to one row, but may be disposed in any number of rows. Regardless of the eyelets sizes, the pattern of eyelets **604** in this commissure feature is symmetrical to the central longitudinal axis of body **602**.

[0068] In another embodiment of commissure feature **600**, illustrated in FIG. 6E, three rows of substantially the same size square eyelets **604** are disposed toward the distal end **620** of body **602** and a single, elongated eyelet **606** is disposed near the proximal end **610** of the body. This commissure feature further includes a pair of recesses **608** disposed on the side edges of body **602** near the distal end **620**.

[0069] FIG. 6F illustrates an embodiment of commissure feature **600** that is similar to that of FIG. 6E, but with recesses **608** disposed near the proximal end **610** of body **602** (adjacent the single eyelet **608**). The eyelets **604** in this commissure feature may be larger or smaller than those in the commissure feature of FIG. 6E.

[0070] FIG. 6G illustrates an embodiment in which commissure feature **600** includes four rows of eyelets **604** and an additional elongated eyelet **608**. It will be understood from this embodiment that the number of rows of eyelets **608** may be varied as desired. Body **602** further includes recesses **608** near its proximal end **610**.

[0071] The commissure feature **600** of FIG. 6H is substantially the same as that of FIG. 6G, but includes addi-

tional recesses **608** on the distal end **620** of body **602** adjacent either side of the strut **614**.

[0072] A commissure feature **600** with a slightly shorter body **602** is illustrated in FIG. 6I. This commissure feature includes three rows of eyelets **604** which are substantially evenly spaced on body **602**. The eyelets **604** in the first row (beginning at the proximal end **610** of body **602**) are slightly larger than those in the second and third rows. They also may be slightly smaller than the eyelets in the other rows. The row of eyelets having a different size (smaller or larger) need not be the first row, but may be any row. Further, the different sized eyelets need not be limited to one row, but may be disposed in any number of rows. Although commissure feature **600** is depicted in FIG. 6I as slightly shorter but about the same width as the commissure features depicted in FIGS. 6A-6H, the commissure feature of FIG. 6I may have a body **602** that is slightly wider than those of the other figures.

[0073] FIG. 6J illustrates yet a further embodiment of commissure feature **600** having three rows of eyelets **604** of substantially the same size and shape disposed on the shorter, body **602**. Additionally, this commissure feature includes a single, elongated eyelet **606** disposed at the proximal end **610** of body **602**, and a pair of recesses **608** disposed on the side edges of the body near its proximal end.

[0074] FIG. 6K illustrates a commissure feature **600** having three dome-shaped bodies **602** connected in series via struts **614**. In this example, each dome-shaped body **602** has a pair of eyelets **604**. The two eyelets **604** may be the same size or different sizes as illustrated. As seen in FIG. 6K, commissure feature **600** includes three bodies **602**, although it will be understood that commissure feature **600** may include two, three, four, five, six or more bodies **602**.

[0075] FIG. 6L illustrates a commissure feature **600** having three dome-shaped bodies **602** connected in series via struts **614** similar to that of FIG. 6K. In this example, however, two of the bodies **602** are connected via suspension struts **624**. Suspension struts **624** may provide greater flexibility for commissure feature **600** during valve loading into a delivery device, in the process of delivery, and in operation of the heart valve. As seen in FIG. 6L, in one example, suspension struts **624** are formed as a pair of struts that diverge at a first body and begin to converge halfway before meeting at a second body **602**. Suspension struts **624** may be formed in the shape of a diamond as shown in FIG. 6L. Alternatively, instead of diamond-shaped struts, struts **624** may form a serpentine spring, flat coil. Bodies **602** may also be disjointed or connected by suture, fabric and/or a polymer. In at least some examples, struts **624** may include radiopaque markers, interlocks for valve-in-valve configurations, or delivery system interlock sites. Additionally, though shown between bodies **602**, struts **624** may be disposed above and/or below each of the bodies.

[0076] As seen in FIG. 6M, the commissure feature need not include three identical dome-shaped bodies, but may include oval bodies **602** having single eyelets. Oval bodies **602** may be disposed in series at the end of the commissure feature **600** or may alternate with other multiple-eyelet bodies in a column. It will be understood that suspension struts **624** may connect any or all of the bodies described herein.

[0077] FIG. 6N illustrates a commissure feature **600** having a dome-shaped body **602** connected to a second body **602** having a square configuration and four eyelets **604**.

Thus, in this configuration, one body includes a single column of eyelets, while a second body includes multiple columns of eyelets **604**. As seen in FIGS. **6N** and **6O**, different-shaped bodies may be connected by single struts **614** or suspension struts **624**.

[0078] Two square-shaped bodies **602** may be connected to form a commissure feature **600** as seen in FIG. **6P**. As seen in this configuration, each body **602** includes four eyelets **604** arranged in two rows and two columns. Suspension struts **624** connect the two square-shaped bodies.

[0079] FIGS. **6Q** and **6R** illustrate two additional examples in which the commissure features include square-shaped bodies connected to rectangular bodies. In FIG. **6Q**, a square-shaped body is connected to a rectangular body **602** having two columns of three eyelets **604**. These two bodies **602** are connected via suspension struts **624**. In the examples shown in FIG. **6R**, the square-shaped body **602** is connected to a rectangular body **602** having two rows, each row having three eyelets **604**.

[0080] FIG. **6S** illustrates a commissure feature having a dome-shaped body **602** connected to a rectangular body having two rows of eyelets, each row having three eyelets **604**. It will be understood that the present invention contemplates various combinations of oval, dome-shaped, square. Moreover, the eyelets may also be circular, rectangular, trapezoidal as well as any other suitable shape.

[0081] Although the commissure features have been described herein as having three or four row of eyelets, any number of rows and/or columns of eyelets may be used, each row and/or column having any number of eyelets. Preferably, the commissure features have at least two rows of eyelets, more preferably at least three rows of eyelets, with at least two eyelets in each row (in other words, at least two columns of eyelets). Moreover, the shapes and sizes of the eyelets in one or more rows may be the same as or different from the shapes and sizes of the eyelets in the other rows. Preferably, however, the shapes and sizes of the eyelets within each row are substantially the same so that the commissure feature has substantial symmetry with respect to the central longitudinal axis of body **602**.

[0082] FIG. **7A** illustrates the previous version commissure feature **500**, while FIG. **7B** illustrates one of the commissure features **600** according to the present invention. Commissure feature **600** is similar to that described above with reference to FIG. **6F** and includes an elongated eyelet **606**, and three rows of square eyelets **604** having substantially the same size, each row having two eyelets. Commissure feature **600** further includes a pair of recesses **608** near the proximal end of body **602** for severally attaching a suture to the body. A silhouette of the end of a leaflet **750** is superimposed over the commissure feature in each figure. As seen by comparing FIGS. **7A** and **7B**, commissure feature **600** has a larger area and more eyelets than commissure feature **500**. As a result, leaflet **750** may be supported by a larger area with commissure feature **600** than with commissure feature **500**. Thus, the load exerted on leaflets **750** may be distributed more uniformly over a larger area with commissure feature **600**, resulting in better performance and less likelihood of failure as compared to commissure feature **500**. Moreover, as will be discussed in greater detail, suture attachment to commissure feature **500** typically results in a terminating knot in a cell of the stent below the commissure feature **500**. Instead, as will be described below, termination of the suture attachment is

disposed in the most distal eyelet within the commissure feature. The top portion of leaflet **750** may be aligned above, below or in-line with the top of the commissure attachment.

[0083] FIG. **7C** illustrates another example of a commissure feature according to the present invention. Commissure feature **700** of FIG. **7C** combines desirable features of both commissure features **500** and **600** described above. Specifically, commissure feature **700** is both shorter and narrower to allow for more flexibility when tracking over a curve. Moreover, when collapsed, commissure feature **700** is less likely to interfere with adjacent struts because it can nest within thin support struts between the aorta and annulus sections without interference. The suture attachment is allowed to terminate in the bottom T-shaped eyelet **706** of commissure feature **700**. Additionally, the leaflet belly suture itself may terminate within the commissure feature so as to protect it during crimping, loading, resheathing, and distortion after implant.

[0084] In addition to the improved commissure feature, the leaflets may be constructed to yield improved performance. For example, the size, shape and curvature of the leaflets may be chosen to correspond to the commissure feature to which the leaflets will be attached. FIGS. **8A-C** illustrates several embodiments of leaflets exhibiting these design considerations.

[0085] FIG. **8A** illustrates a previous version of a leaflet **750** having a free edge **752** for coapting with other leaflets and a tab **754** at each end of the free edge for attaching the leaflet to a commissure feature. In some examples, two leaflets may be attached to each other and to a commissure feature.

[0086] FIG. **8B** illustrates a leaflet **850** according to the present invention. Leaflet **850** has a free edge **852** for coapting with other leaflets and a tab **854** at each end of the free edge for attaching the leaflet to a commissure feature. As seen by comparing the leaflets in FIGS. **8A** and **8B**, the dimensions and shapes of a leaflet may be modified to improve performance. Specifically, the tab **754** of leaflet **750** is formed with height $y1$ and width $x1$. While this height and width may be sufficient to fully cover the previous version commissure feature **500**, they may be too small to fully cover the longer commissure features **600** of the present invention. In contrast, the tab **854** of leaflet **850** may have an increased height of $y2$, enabling the tab to fully cover, and even extend beyond the height of, commissure feature **600**. Tab **854** may also have a greater width $x2$ than tab **754**. This greater width provides tab **854** with a greater area to fold about fold line **856** and attach to a commissure feature **600**, thereby providing leaflet **850** with greater abrasion resistance. The height of the belly portion of the leaflets may also be modified. As seen in FIG. **8A**, leaflet **750** has a belly height of $f1$ between the bottom of the leaflet and the free edge **752**. In contrast, the belly height $f2$ of leaflet **850** may be greater. Without being bound by any particular theory, it is believed that the greater belly height of leaflet **850** may allow for better leaflet coaptation in elliptical and other distorted configurations.

[0087] FIG. **8B** further illustrates the use of an additional reinforcement material **860** being used as a stress distributor, reinforced and buffer. A thin layer of reinforcement material **860** may be sandwiched and/or disposed between the folded layers of tab **854**. The tab **854** may be folded over the reinforcement material **860** to provide additional strength and support to the leaflet-commissure attachment area. The

reinforcement material **860** may be a fabric, tissue or polymer sheet material and may be attached via sutures, bio-glue, weaving and or slots in the reinforcement material.

[0088] As seen in FIGS. **8C** and **8D**, additional material may be attached to tabs **854** in the tabs may be formed with additional structures. For example, in FIG. **8C**, each tab **854** includes an additional flap **858** and an additional fold line **856**. Flaps **858** may be formed unitarily with tabs **854** or separately and then attached to the tabs. Flaps **858** may be formed from the same material as tabs **854** or join a different material. The additional flap **858** may be disposed on the top of tab **854**, as shown in FIG. **8D**. In addition, leaflet **850** may have a single elongated tab **854** on only one side of the leaflet, the elongated tab **854** being wide enough to wrap over an end or tab of the next adjacent leaflet before attaching to a commissure feature.

[0089] The size and shape of the free edge of the valve leaflets may also be modified to further improve performance. As seen in FIG. **9A**, free edge **752** of leaflet **750** has a curved portion **c1** which attaches to tab **754** at junction **910**. When attached to a commissure feature, especially a larger commissure feature **600**, such a configuration may limit the performance and movement of free edge **752**. A leaflet **850** according to the present invention may be formed as illustrated in FIG. **9B**. Leaflet **850** may include a free edge **852** having a curved portion **c2** coupled to tab **854**. Tab **854** may include a substantially vertical edge **855** having a height **v1**. Curved portion **c2** may attach to tab **854** at junction **910** adjacent vertical edge **855**. Providing a vertical edge **855** on tab **854** may facilitate suturing of the tab to the commissure feature without interfering with the function of the free edge **852** of the leaflet.

[0090] In addition to modifying the commissure feature and the leaflet configuration, the suture pattern attaching the leaflets to the commissure features may also be modified to improve performance. One example of a modified suture pattern according to the present invention is illustrated in FIGS. **10A** and **10B**. FIG. **10A** illustrates the leaflet-commissure feature attachment as seen from the abluminal side of a valve assembly, while FIG. **10B** illustrates the same attachment as seen from the luminal side of the valve assembly.

[0091] As seen in FIG. **10A**, a commissure feature **600** includes a body **602** having a proximal end **610** and a distal end **620** similar to the examples described in FIGS. **6A-J**. Commissure feature **600** may include a plurality of eyelets **604** disposed toward the distal end **620** of body **602** and a single, elongated eyelet **606** disposed near the proximal end **610** of the body. Commissure feature **600** may also include a pair of recesses **608** disposed on the side edges of body **602** near the proximal end **610** thereof. The plurality of eyelets may include six substantially square eyelets **604** arranged in three rows with two eyelets in each row. Eyelets **604a** and **604b** are disposed in the first row of eyelets, beginning at the proximal end **610** of body **602**. Eyelets **604c** and **604d** are arranged in the middle row of eyelets, and eyelets **604e** and **604f** are arranged in the most distal row of eyelets.

[0092] As seen in FIG. **10B**, the tabs **854** of two different leaflets **850** are attached to commissure feature **600** using a suture or multiple sutures. The following describes the use of, a single suture **S** to attach leaflets **850** to commissure feature **600**. It will be understood, however, that multiple sutures may be used for this purpose. For example, one suture may attach a first leaflet **850** to commissure feature

600, while a second, separate suture attaches the second leaflet **850** to the commissure feature.

[0093] The use of a single suture to attach two leaflets **850** to a commissure feature **600** will be described with reference to FIG. **10C**, which shows an enlarged view of the leaflet-commissure assembly from the abluminal side of the valve assembly and depicts the suture pattern through several positions **p1-16** on the assembly. It will be appreciated that FIG. **10B** shows the corresponding suture pattern from the luminal side of the valve assembly and that the suture pattern may be understood from this view as well.

[0094] The suture pattern may begin at any point at or near commissure feature **600** and terminate at any other point. In at least some examples, the suture pattern begins and terminates at the same position. For the sake of illustration, the suture pattern will be described as beginning at point **P1**, within eyelet **604b**. As used herein, with reference to FIG. **10C** and other figures showing the abluminal side of the valve assembly, the term “out” indicates passing the sutures **S** from the luminal side of the valve through the leaflet tab **854** and past the stent structure to the abluminal side of the valve while the term “in” indicates passing the suture from the abluminal side of the valve past the stent structure and through the leaflet tab **854** to the luminal side of the valve.

[0095] The suture pattern **S** may begin by passing suture **S** out through eyelet **604b** at point **p1**. Suture **S** may then be advanced in through point **p2** in eyelet **604d**, back out through point **p3** in eyelet **604b**, and finally in through point **p4** in eyelet **604d**, essentially forming two loops of suture **S** around rib **651**. Suture **S** may then be directed up toward eyelet **604f** and passed out through point **p5** in eyelet **604f** adjacent central spine **655**, and then in through point **p6** in eyelet **604e** adjacent the opposite side of the central spine subsequently, suture **S** may be passed out through point **p7** in eyelet **604c** in through point **p8** in eyelet **604a**, back out through point **p9** in eyelet **604c**, and back in through point **p10** in eyelet **604a**, essentially forming two loops of sutures around rib **657**. Suture **S** may then be directed down toward elongated eyelet **606** and passed out through point **p11** in eyelet **606**, over the side edge of body **602** at point **p12** positioned within recess **608**, back out through point **p13** in eyelet **606**, in through point **p14** in the same eyelet, around the side edge of body **602** at point **p15** within the second recess **608**, and in through point **p16** in eyelet **606**. The tail (not shown) of suture **S** at point **p16** may then be knotted, tied or otherwise joined to the tail (not shown) of the suture at point **p1** to complete the attachment of commissure feature **600**.

[0096] As seen in FIG. **10C**, two optional loops **L1** and **L2** may be used to tie the leaflets **850** to the commissure feature **600** at the distal end **620** of body **602**. Loops **L1** and **L2** may be formed from one or more lengths of suture separate from suture **S**. Loop **L1** may begin at point **p17** in eyelet **604e**, and loop over the top of body **602**, with the ends of the loop tied together. Loop **L2** may form a similar attachment at point **p18** in eyelet **604f**. As described above with particular reference to FIG. **6H**, body **602** may have recesses formed on the distal end **620** of body **602** to aid in securing the loop sutures.

[0097] Another example of a suture pattern according to the present invention is illustrated in FIGS. **11A** and **11B**. FIG. **11A** illustrates the leaflet-commissure feature attachment as seen from the abluminal side of a valve assembly,

while FIG. 11B illustrates the same attachment as seen from the luminal side of the valve assembly.

[0098] As seen in FIG. 11A, a commissure feature 600 has a body 602 with a proximal end 610 and a distal end 620 similar to the examples described in FIGS. 6A-J. Commissure feature 600 may include a single, elongated eyelet 606 disposed near the proximal end 610 of body 602 and six substantially square eyelets 604 disposed toward the distal end 620 of the body, the eyelets being arranged in three rows of two eyelets each. Eyelets 604a and 604b are disposed in the first row of eyelets, beginning the proximal end 610 of body 602. Eyelets 604c and 604d are arranged in the middle row of eyelets, and eyelets 604e and 604f are arranged in the most distal row of eyelets. A pair of recesses 608 may be provided in the side edges of body 602 near the proximal end 610 thereof.

[0099] As seen in FIG. 11B, the tabs 854 of two different leaflets 850 are attached to commissure feature 600 using a sutures. The suture pattern depicted in FIG. 11A is similar to the pattern described above with reference to FIGS. 10A-C. In addition to the horizontal stitches (as seen in FIG. 11A) within and around elongated eyelet 606 and the vertical stitches (as seen in FIG. 11A) connecting eyelet 604a to eyelet 604c and eyelet 604d to eyelet 604b, the suture pattern incorporates the two most distal eyelets 604e and 604f using an “X” pattern. More particularly, suture S may pass out through eyelet 604f at point p1, travel diagonally over the central spine 655 of body 602 and in through point p2 in eyelet 604e, travel horizontally (as seen in FIG. 11A) under the central spine and back out through point p3 in eyelet 604f, and then travel diagonally over the central spine again (in the opposite orientation) and in through point p4 in eyelet 604e, creating an “X” pattern on the abluminal surface of the valve assembly. It will be understood that this portion of the pattern may be incorporated as part of a single suture pattern coupled to the suture pattern in the other eyelets 604a-d and/or in the elongated eyelet 606, or may be formed as a separate suture pattern and tied together.

[0100] Another example a suture pattern according to the present invention is illustrated in FIGS. 12A and 12B. FIG. 12A illustrates the leaflet-commissure feature attachment as seen from the abluminal side of a valve assembly, while FIG. 12B illustrates the same attachment as seen from the luminal side of the valve assembly.

[0101] The pattern of suture S illustrated in FIGS. 12A and 12B is similar to that described above with reference to FIGS. 11A and 11B, except that the distal “X” pattern is disposed on the luminal side of the valve assembly as opposed to on the abluminal side of the assembly. When viewed from the abluminal side of the assembly the suture pattern merely reveals a horizontal stitch between eyelets 604e and 604f.

[0102] A suture pattern according to the present invention disposed on a commissure feature 600 having only six eyelets is illustrated in FIGS. 13A and 13B. FIG. 13A illustrates the leaflet-commissure feature attachment as seen from the abluminal side of a valve assembly, while FIG. 13B illustrates the same attachment as seen from the luminal side of the valve assembly.

[0103] The suture pattern, shown in FIGS. 13A and 13B is similar to that described above with references to FIGS. 11A and 11B, without the proximal portion of the pattern incorporating elongated eyelet 606. It will be understood that the patterns described herein are merely exemplary and that

combinations of these patterns may be made as desired depending on the particular structure of the commissure feature being used. Moreover, in addition to the overall suture pattern, the order of the pattern may also be modified. For example, the commissure attachment feature may be assembled clockwise or counter-clockwise, front to back, back to front, left to right, right to left, or eyelets may be skipped and then later attached. This allows for different varying levels of security, stress concentrations, abrasion points as well as symmetrical and non-symmetrical attachments.

[0104] Although the invention herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present invention. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined by the appended claims. It will be appreciated that the various dependent claims and the features set forth therein can be combined in different ways than presented in the initial claims. It will also be appreciated that the features described in connection with individual embodiments may be shared with others of the described embodiments.

1. A prosthetic heart valve, comprising:

- a collapsible and expandable stent having an inflow end and an outflow end, the stent including a plurality of struts defining cells and a plurality of commissure attachment features, each of the plurality of commissure attachment features including a body having a proximal end, a distal end and a plurality of eyelets, the stent extending in a length direction from the inflow end to the outflow end;
- a collapsible and expandable valve assembly including a plurality of leaflets connected to the plurality of commissure attachment features, each of the plurality of leaflets having a first tab and a second tab; and
- a suture pattern coupling first tabs of two adjacent leaflets of the plurality of leaflets to a first commissure attachment feature of the plurality of commissure attachment features, wherein the suture pattern passes through each of the plurality of eyelets and includes at least one vertical segment and at least one horizontal segment.

2. The prosthetic heart valve of claim 1, wherein the plurality of eyelets includes first-sized eyelets arranged in at least two rows extending in a circumferential direction of the stent and at least two columns extending in a length direction.

3. The prosthetic heart valve of claim 2, wherein the plurality of eyelets further includes an elongated eyelet that is wider than the first-sized eyelets.

4. The prosthetic heart valve of claim 3, wherein the first-sized eyelets and the elongated eyelet are all symmetrically disposed about a center longitudinal axis extending in the length direction from the proximal end of the body to the distal end of the body.

5. The prosthetic heart valve of claim 2, wherein each of the first-sized eyelets is of a same size.

6. The prosthetic heart valve of claim 3, wherein the body of each commissure attachment feature has a peripheral edge and at least one recess formed in the peripheral edge.

7. The prosthetic heart valve of claim 6, wherein the at least one recess is aligned in a circumferential direction of the stent with elongated eyelet.

8. The prosthetic heart valve of claim 6, wherein the at least one recess of the peripheral edge is aligned in a circumferential direction of the stent with the elongated eyelet.

9. The prosthetic heart valve of claim 8, wherein two recesses are formed in the peripheral edge of each commissure attachment feature, each of the recesses being aligned in the circumferential direction of the stent with the elongated eyelet.

10. The prosthetic heart valve of claim 8, wherein the peripheral edge of each commissure attachment feature includes a proximal edge on the proximal end of the body, a distal edge on the distal end of the body, and a pair of side edges between the proximal edge and the distal edge, and the at least one recess is disposed on one of the side edges of the body.

11. The prosthetic heart valve of claim 3, wherein the suture pattern terminates in a knot positioned at least partially within the elongated eyelet.

12. The prosthetic heart valve of claim 2, wherein the at least one vertical segment extends between the first-sized eyelets.

13. The prosthetic heart valve of claim 2, wherein the at least one vertical segment includes multiple vertical segments.

14. The prosthetic heart valve of claim 2, wherein the at least one vertical segment includes vertical segments that wrap around an upper portion of the body of first commissure attachment feature.

15. The prosthetic heart valve of claim 3, wherein the at least one horizontal segment is disposed adjacent the elongated eyelet.

16. The prosthetic heart valve of claim 9, wherein the suture pattern is received within the two recesses on the peripheral edge of each commissure attachment feature.

* * * * *